Quobly in June 2026

Headquarters in Grenoble, France
Spin-off in 2022 from cea cnrs



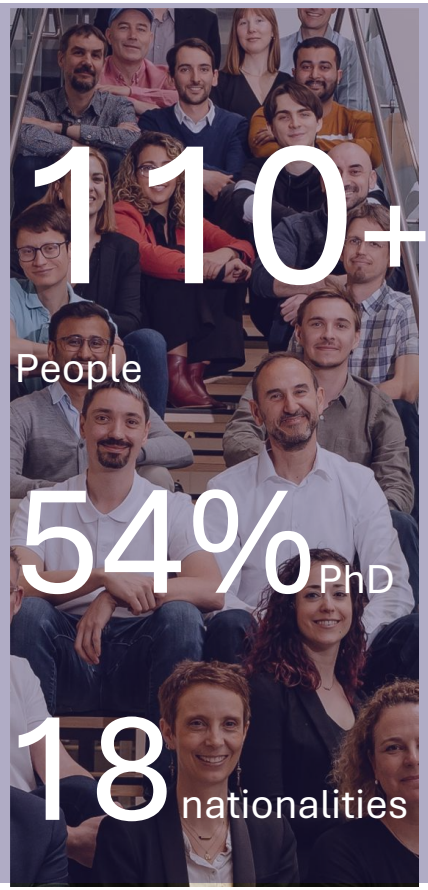
2023 EIC Transition Awardee

2024 EY Entrepreneur of the year

2025 EIC Acceleration Awardee

Class of 2025 French Tech 2030

65 patents



15+ academic partnerships



Offices in **Paris** & **Taiwan**

Subsidiaries in **Singapore & Canada**



Stakeholders: QUANTONATION bpifrance SUPER NOVA INV INNOVACOM

Our networks: IEC ISO MINALOGIC EIC ScalingClub Selected Club Member le lab quantique QUIC QuantAlps #QEI QUANTUM FLAGSHIP FRANCE 2030 LES MAISONS DU QUANTIQUE FRANCE DEEPTECH CREATIVE LAB Le HUB Quantique

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Quobly secures €115 million Series A to bring silicon-based quantum computers to market

This financing marks an important milestone for Quobly as we move from technology validation to industrial execution

WHAT WE ACHIEVED
Seed Phase

- Demonstrated feasibility of manufacturing silicon qubits using semiconductor industry processes
- Built a full-stack architecture combining hardware, control electronics and software
- Prepared the deployment of our first commercial quantum computer



WHAT WE WILL DO
With Series A

- Strengthen the performance and scalability of our quantum computing platform
- Accelerate the industrialization of silicon quantum processors
- Deploy first Alloy systems in cloud and HPC environments

Alloy Pioneer — First commercial quantum computer accessible through the cloud in 2026

quobly's edge: semiconductor industry to disrupt Quantum Computing market





10m² quantum computer compatible with HPC and data centers for pervasive hybrid quantum-classical computing

1,000x faster, **100x** smaller, **100x** cheaper

Quantum computing future depends on technology's ability to scale the number of qubits at a competitive cost: Quobly leverages the billions of dollars invested in the semiconductor industry.

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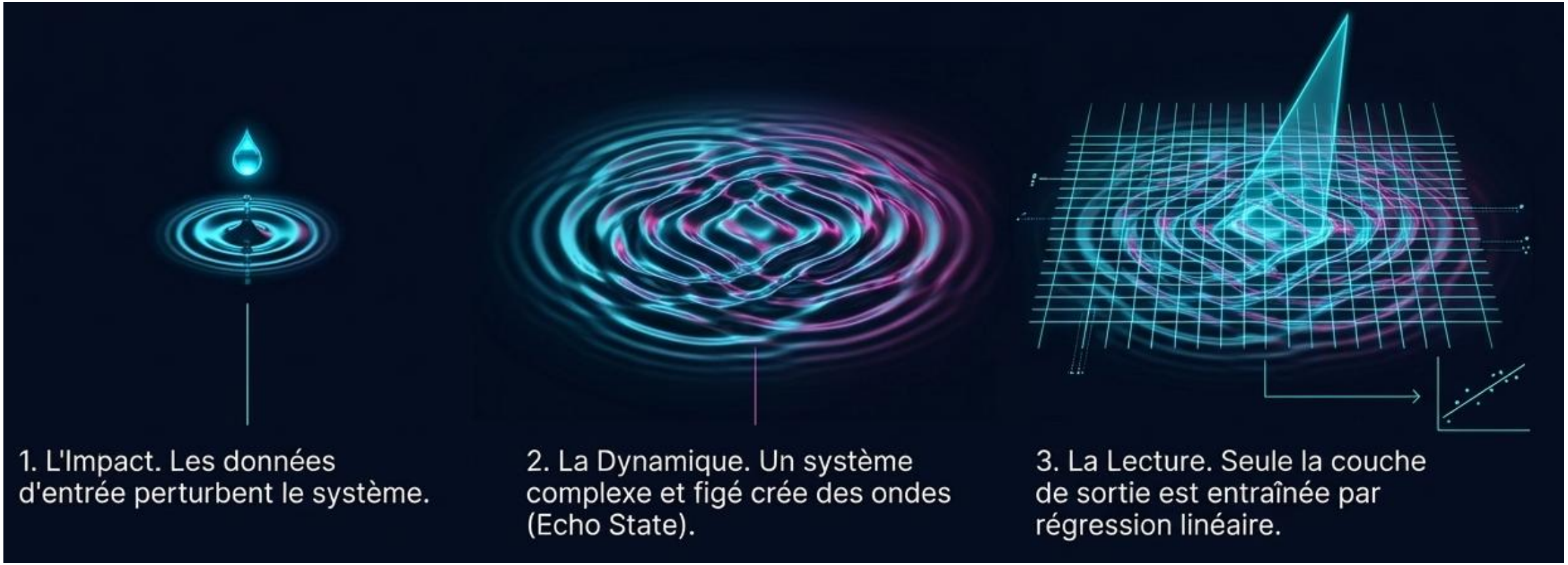
A gentle introduction to « Quantum Reservoir Computing » (QRC)

Complexity as a Resource

Alain Champenois
June 2026 v1.0

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The elegance of RC: freeze the network, train the readout

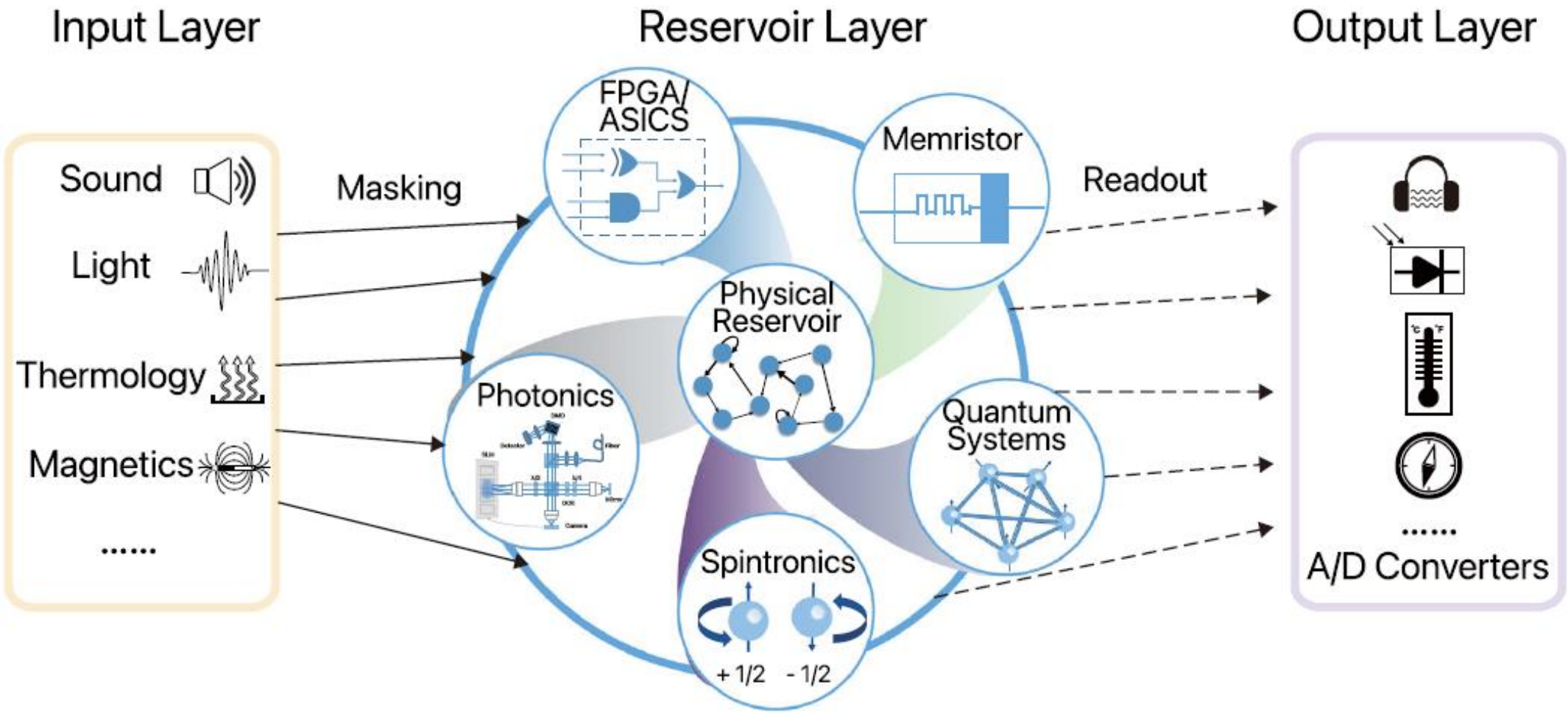


Water Bucket Analogy: Imagine a bucket filled with water. If you throw pebbles into it (your input data), the surface creates complex ripples. The ripples encode the history of the thrown pebbles. By simply observing the surface (**the readout**), you can extract incredible information.

Think of the reservoir as a fixed **non-linear non-Markovian (with memory) dynamical system**. We inject a time-series stream into it, it naturally scrambles the history into a highly expressive state, and we observe what happens. By offloading the heavy computational burden onto the intrinsic physics of this **reservoir**, we only need to add a lightweight classical machine learning model at the readout layer to decode and map the final predictions.

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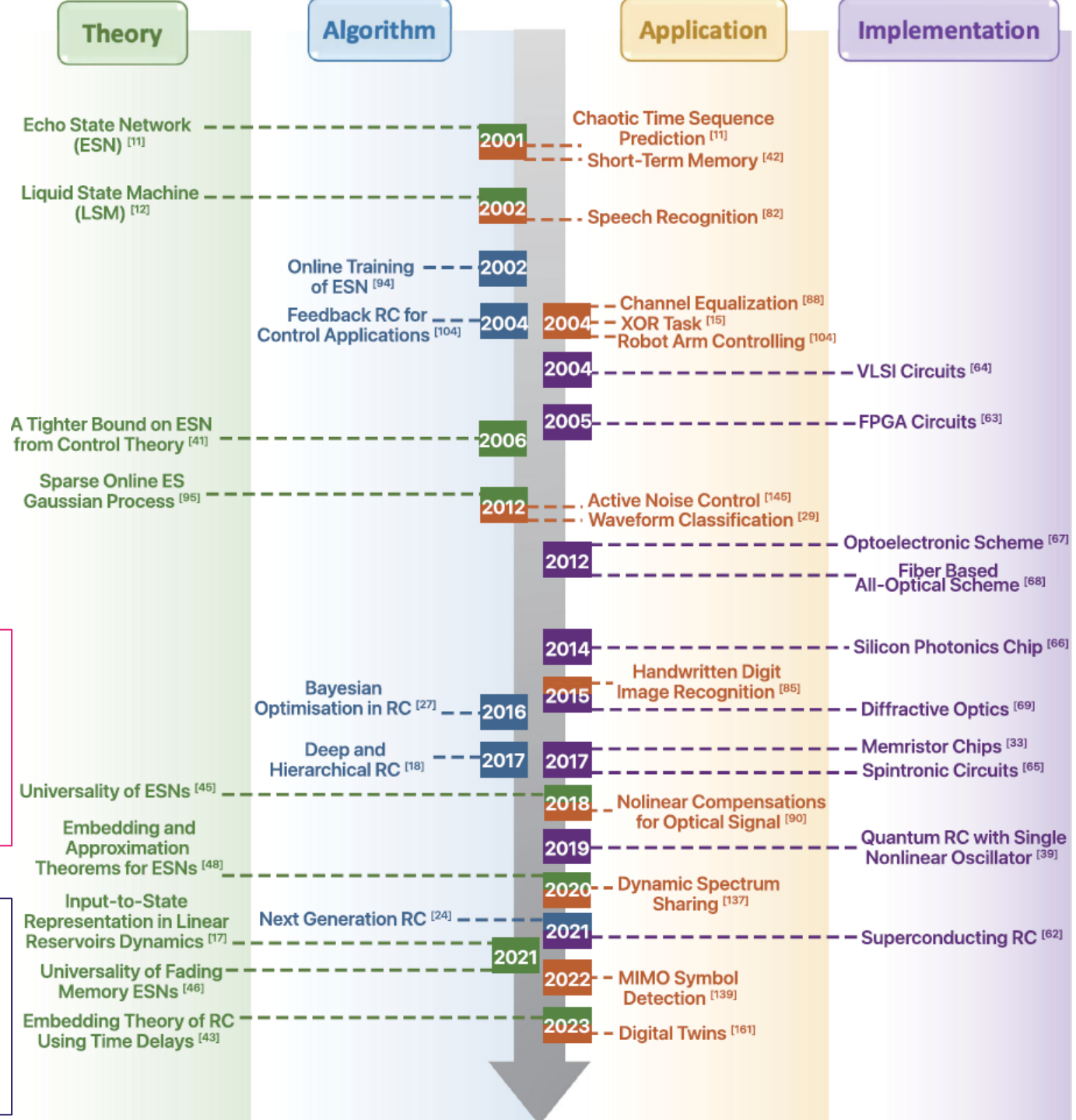
Physical Reservoir Computing (RC)



Emerging opportunities and challenges for the future of reservoir computing (2024)
https://doi.org/10.1038/s41467-024-45187-1

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Historic of RC



Nakajima, **Fuji**: popularization and structuring of the **QRC** approach as an extension of RC into the quantum domain 2017

Had the intuition that the natural dynamics of a quantum system possessed all the properties required to replace an RNN.

They have demonstrated that Quantum RC :

- Can **approximate** any non-linear function of the input history.
- Works even if the reservoir is not trained.
- **Local observables** were sufficient to extract the full richness of the reservoir.

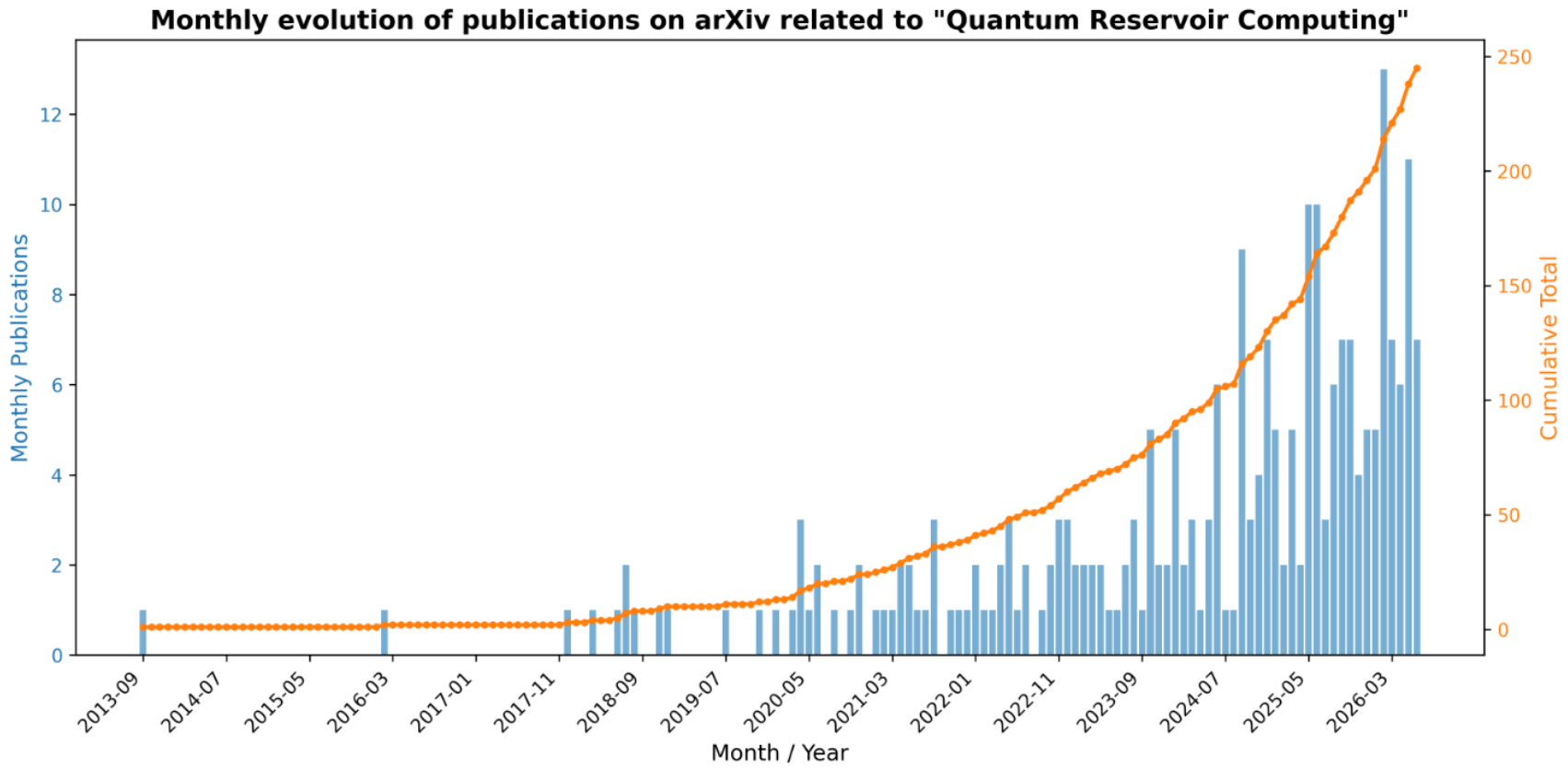
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Scientific Papers on QRC & Quantum Extreme Learning Machine QELM

The proof of universality for Quantum Reservoir Computing (QRC) using local observables was primarily established by **Keisuke Fujii** and **Kohei Nakajima** in their seminal **2017** paper:

« **Harnessing Disordered Quantum Dynamics for Machine Learning (2017)**».

The core idea is that a rich and disordered quantum system can serve as a reservoir, where local observables, when linearly combined, are sufficient to approximate complex, non-linear temporal dynamics.

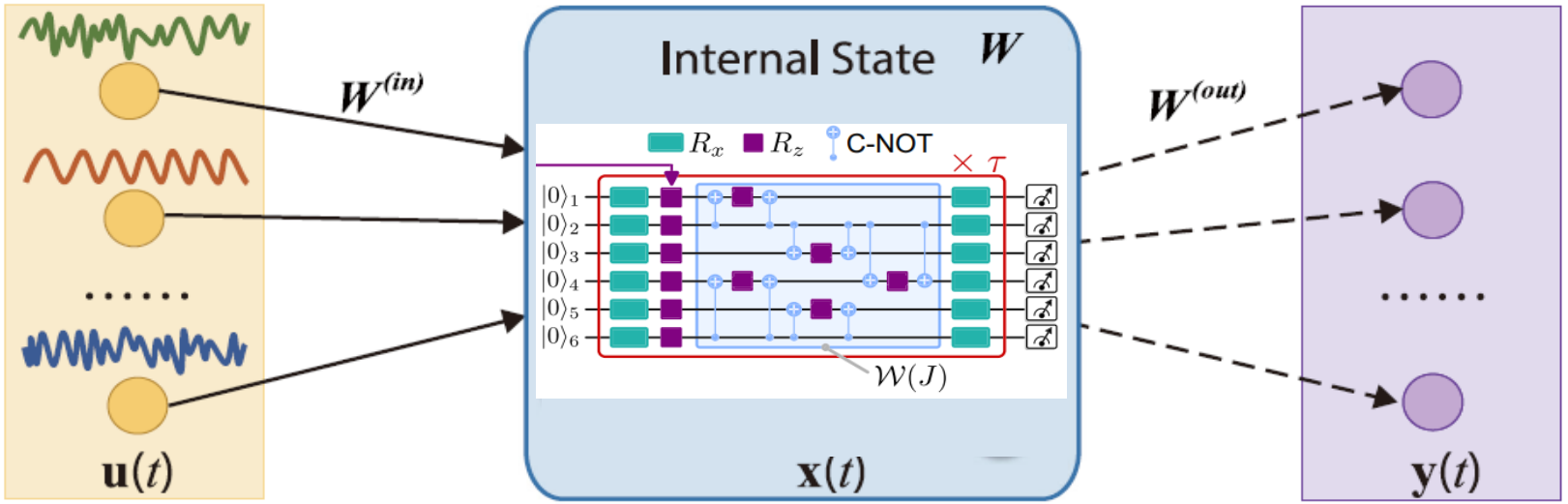


Architecture of the Quantum Reservoir Computing (QRC)

Classical
Input Layer

Quantum
Reservoir Layer

Classical
Output Layer



N qubits

M observables

Batch of measurement shots

Legend:

Fixed Coupling

Trainable Coupling

→ Final Step (READOUT) via a **classical ML Regression** :

This is where the final error is minimized via a regularization on the output weights.

The model's output equation then becomes a linear combination of the observables :

$$y(t) = \sum_{i=1}^{M} w_i \langle \hat{O}_i \rangle_{synth}$$

Where **M** (the number of synthetic observables) can be much larger than the number of physical **qubits N**

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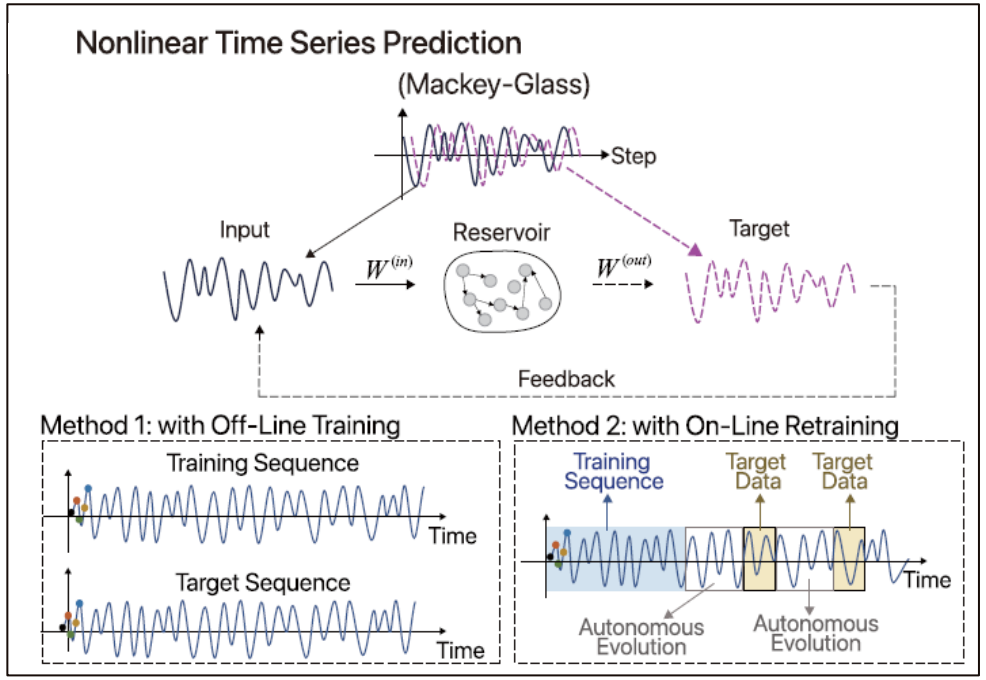
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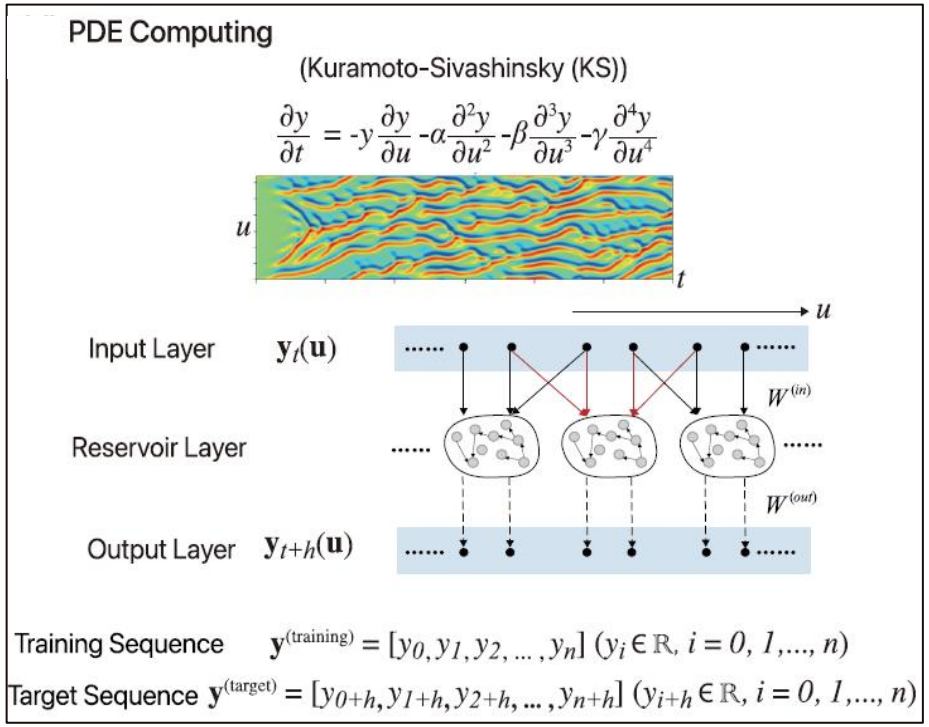
Industrial Applications

QRC for complex non-linear dynamical systems

For non-linear time-series prediction we exploit the **kinematics** of the **Hilbert space** on a fixed topology in the following use-cases :

- **Chaotic temporal systems** where standard classical alternative methods collapse (LSTMs, GRUs, ... with explosive gradient problem)
- **Dynamical systems and fluids dynamics :** weather forecasting & atmospheric dynamics, ocean waves, epidemiological & population dynamics where variables are linked by coupled partial differential equations (PDE).

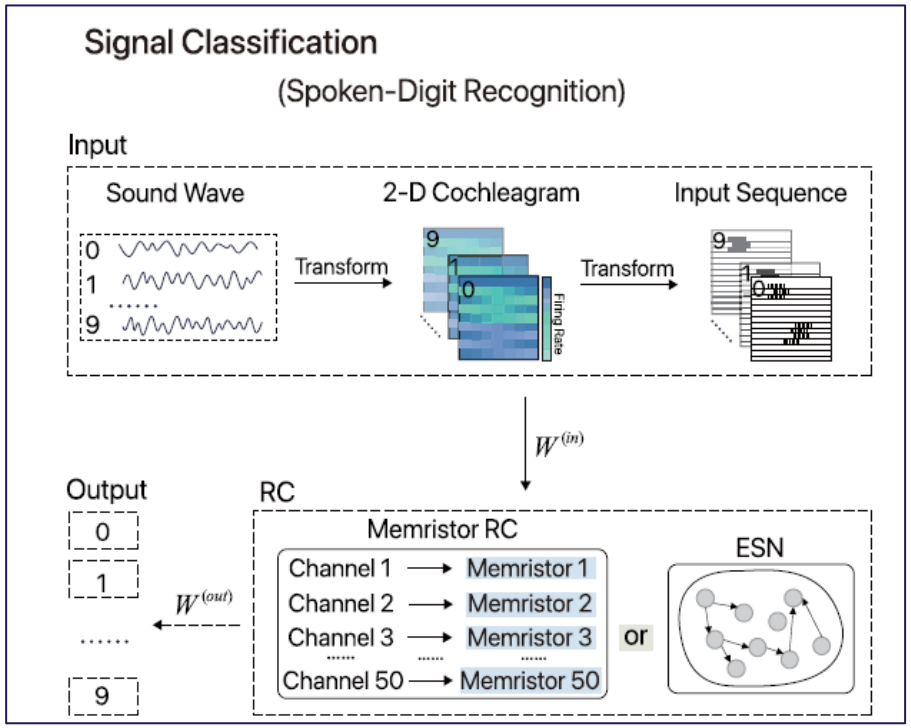




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Industrial Applications of QRC in Dynamical Systems

- "**Soft Sensors**" for **chemical engineering** : some hidden variable cannot be measured directly but by feeding other visible/measured variables we can reconstruct the internal dynamics of the fluid phase to instantly predict the hidden variable like a *virtual sensors*.
- **Temporal signal processor:** speech recognition, anomaly detection in power grids (Smart Grids), financial series, or non-linear time series prediction.



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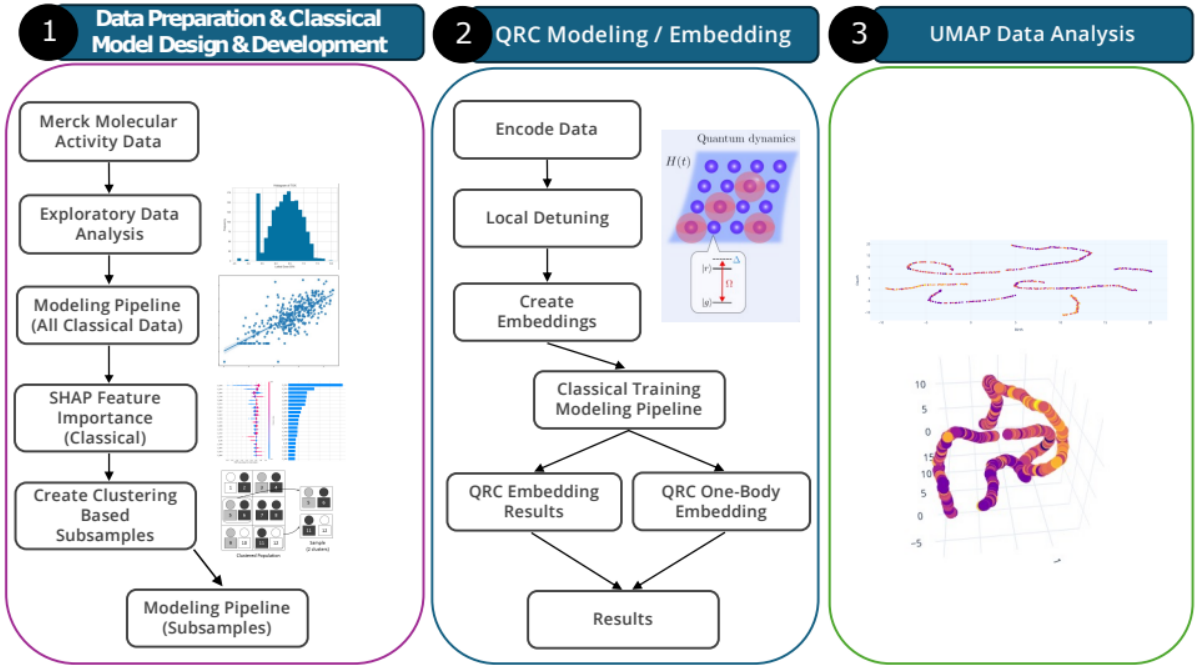
QRC for Classification on Static data

For signal classification we exploit the **geometry** of the **Hilbert space** for **dimensionality** and **separability advantage.**

- For the case of **static data** (Satellite image, sound …), QRC face extremely performant classical alternatives (XGBoost, GNN or Random Forests).
- Data-Scarce Efficiency : in the case of data scarcity (e.g. Molecular descriptors), QRC seems to retain a powerful generalization property with fewer records outperforming or **matching** classical algorithms like **SVMs**.



Breaking concentration barriers for quantum extreme learning on digital quantum processors
https://arxiv.org/abs/2603.13005 Mars 2026)

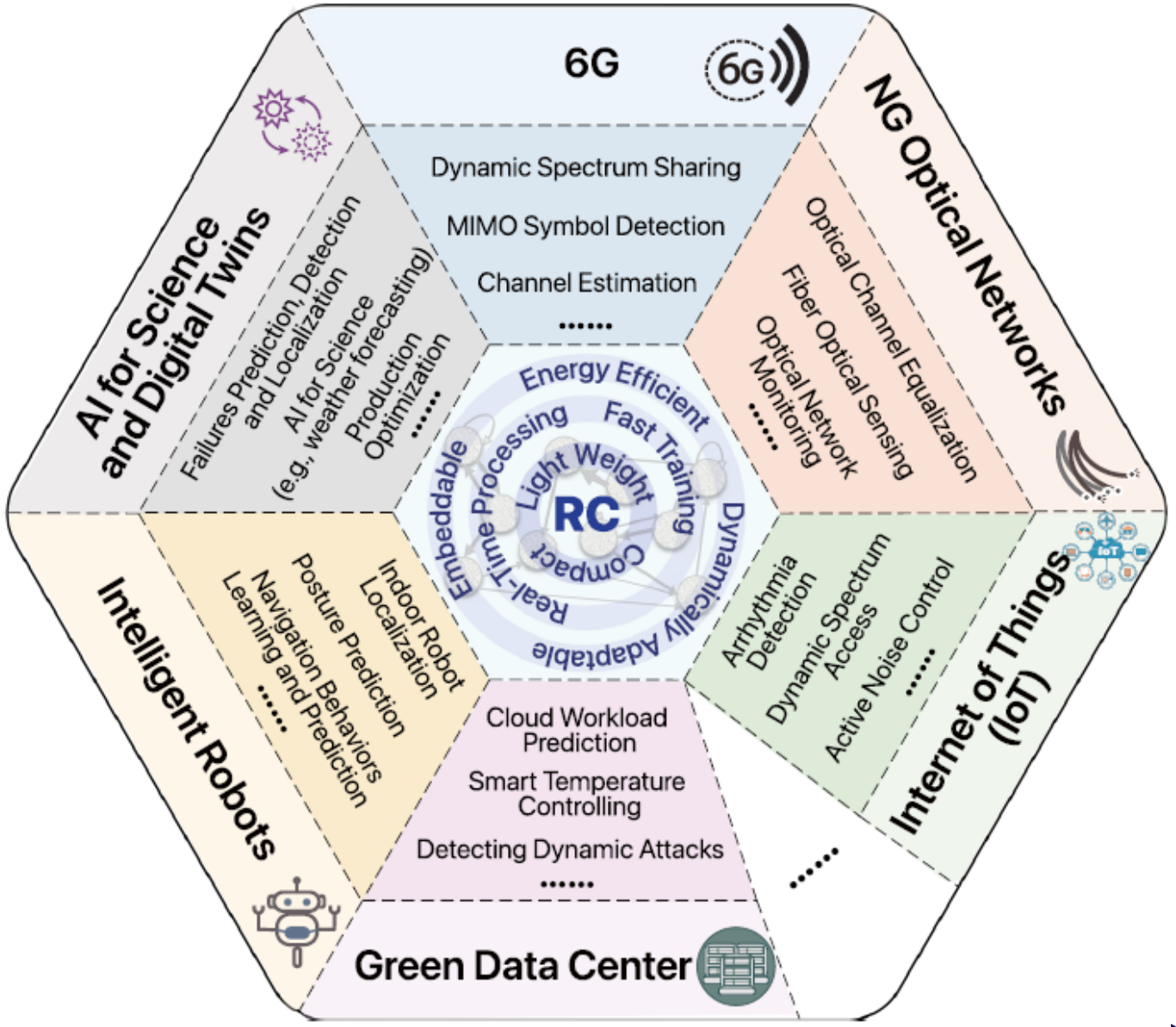


Robust Quantum Reservoir Computing for Molecular Property Prediction
https://arxiv.org/abs/2412.06758

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Application domain where QRC could bring value

Emerging opportunities and challenges for the future of reservoir computing :
https://doi.org/10.1038/s41467-024-45187-1



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Foundation Models & Machine Learning Theory

Towards Geometric Quantum Machine Learning (QML)

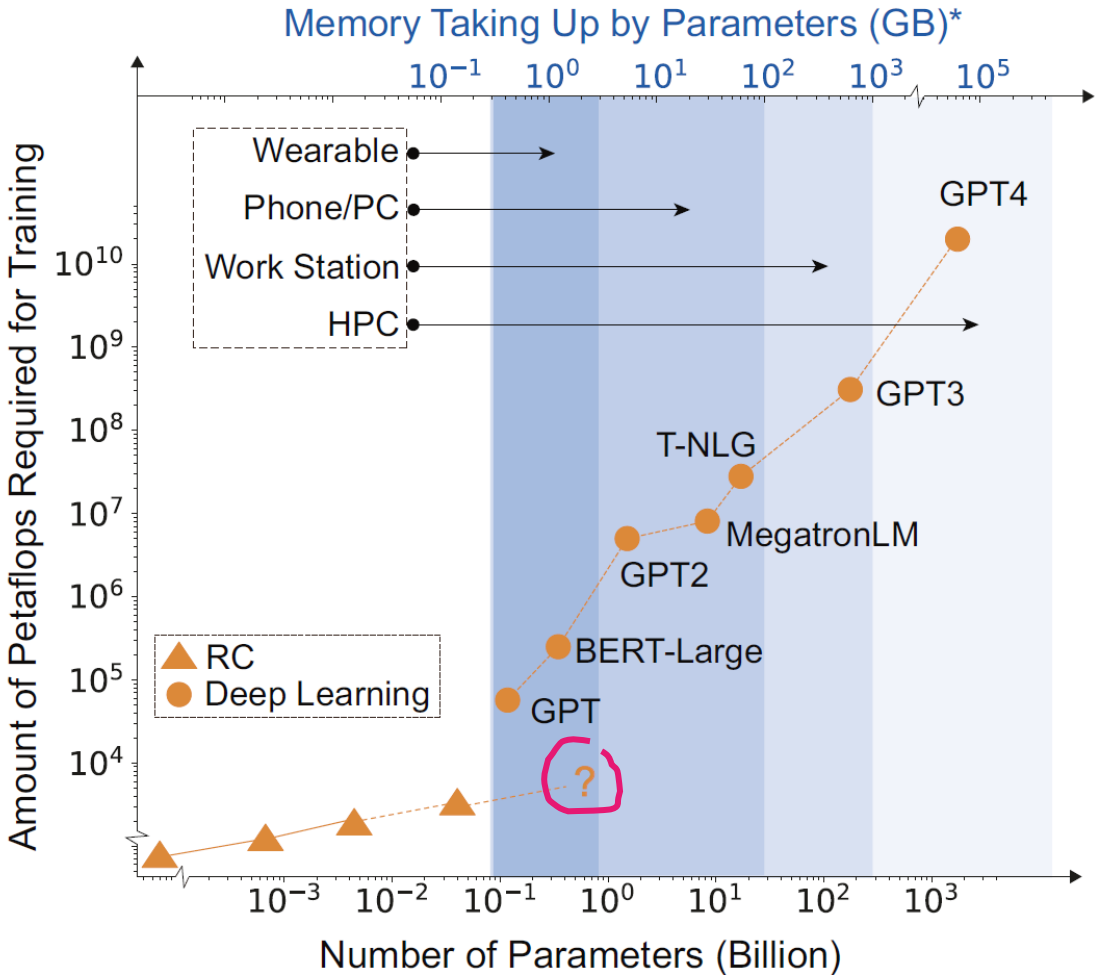
- Except for simple problem but the myth of the ***completely arbitrary*** quantum reservoir, thrown randomly at a complex problem and capable of solving everything by magic is no longer valid : the Hilbert space becomes a stochastic labyrinth with universal mediocrity.

- **The design of a reservoir is no longer arbitrary**: it is engineered to be isomorphic to the mathematical structure of the family of problems being addressed. This is referred to as **Hamiltonian Engineering** or **symmetry-preserving** (equivariant) **QRC.** It refers to mathematical symmetries especially for processing structured data of variable size or shape.

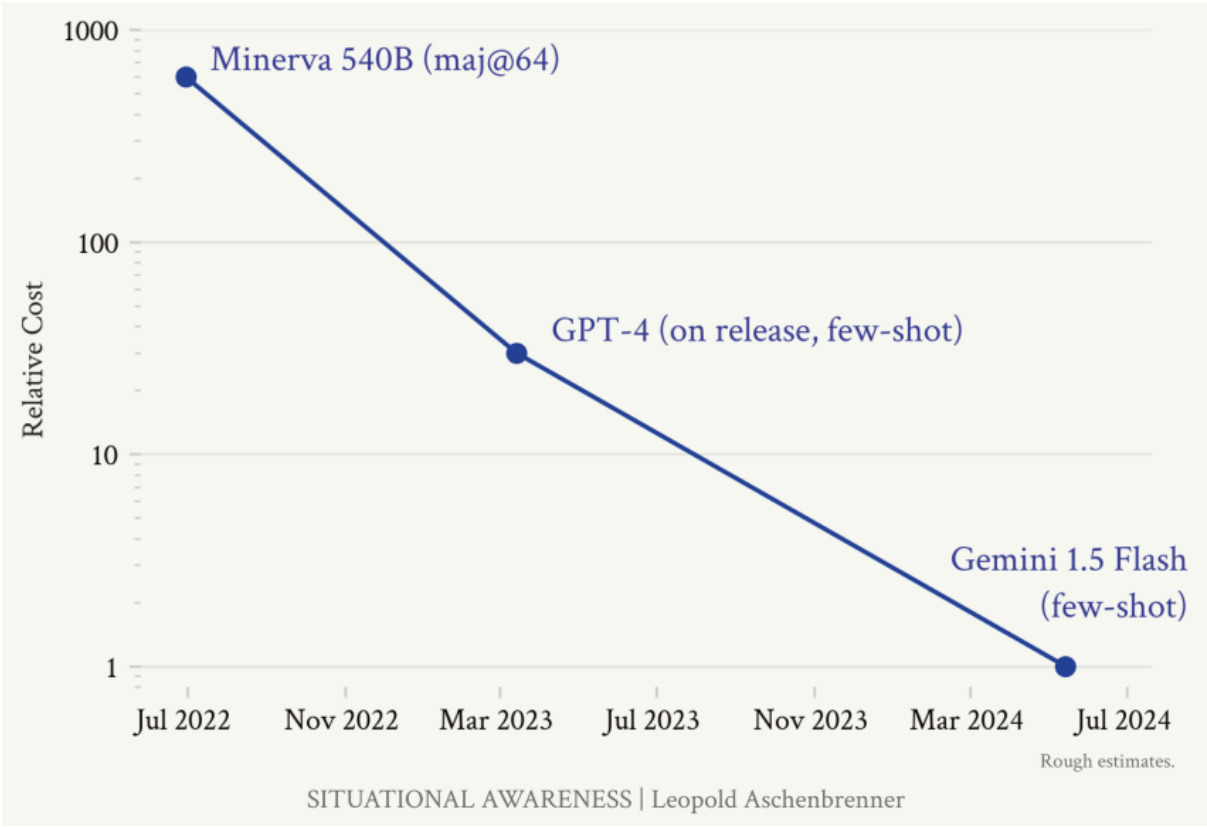
- **Multi-tasking is not dead**, it has changed scale : QRC remains a multi-tasking algorithm, but within a defined mathematical family who share the same geometric structure.

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Architecture Reservoir Computing (QRC) vs Generative AI



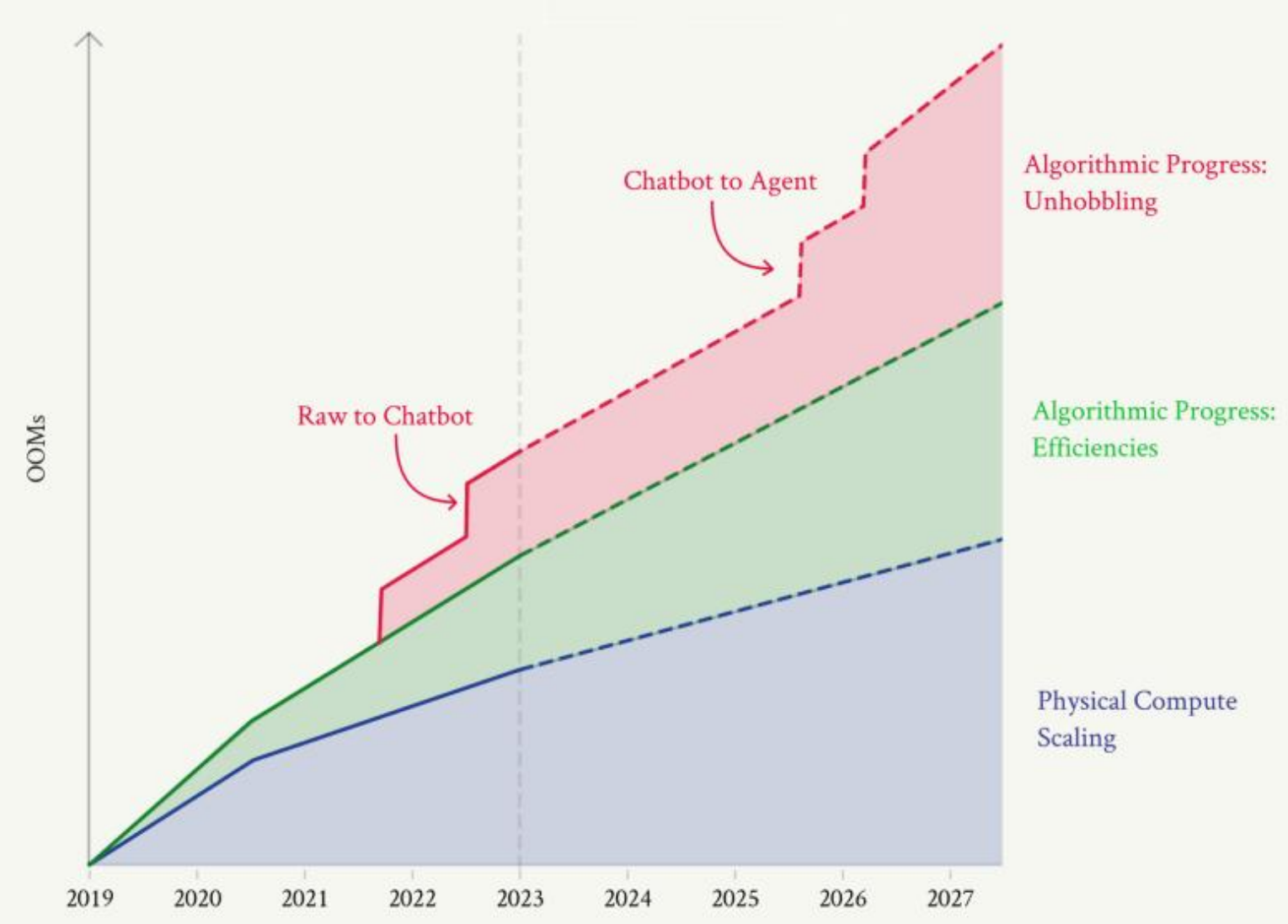
1000x Inference efficiency jump for a MATH benchmarks Score



https://situational-awareness.ai/wp-content/uploads/2024/06/situationalawareness.pdf

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Unlocking Latent Power from complexity ressource



Decomposing progress: compute, algorithmic efficiencies, and unhobbling.

Raw base models hold massive latent intelligence but are notoriously hard to control. **"Unhobbling"** to unleash this dormant capability (~100x computation efficiency) :

- **Reinforcement Learning :** apply its capabilities toward responding to a specific question in a helpful manner.
- **Chain of Thought (CoT) :** use reasoning tasks and thinking for a long time.
- **Scaffolding :** multi-agent architectures and conversation between LLMs.

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End-to-end Quantum Machine Learning (QML) workflow



- Concept of **Quantum embedding** aim to represent **classical data** using **quantum states**. This approach preserves the inherent relationships and structures within the data.

- The **choice** of encoding method largely depends on the **nature of the data** and the specific quantum machine learning task at hand. Each approach brings its own set of advantages and challenges, and understanding their nuances is crucial.

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Comparison of common quantum data encoding strategies

Each strategy balances different **trade-offs** in terms of qubit requirements, circuit complexity, noise resilience, and applicability to data types.

Encoding strategy	Data type	Qubit requirement	Circuit depth	Noise resilience	Example gates
Basis encoding	Binary strings	*n*	Low	High	X, I
Angle encoding	Real-valued features	*n*	Low	Moderate	$R_y(x_i)$, $R_z(x_i)$
Amplitude encoding	Normalized real vector $x \in \mathbb{R}^d$	$\log_2 d$	High	Low	State preparation circuit
Hamiltonian-based encoding	Structured inputs (e.g. graphs, molecules)	Problem-dependent	Structure embedded in entanglement pattern	Moderate	$e^{-iH(x)t}$

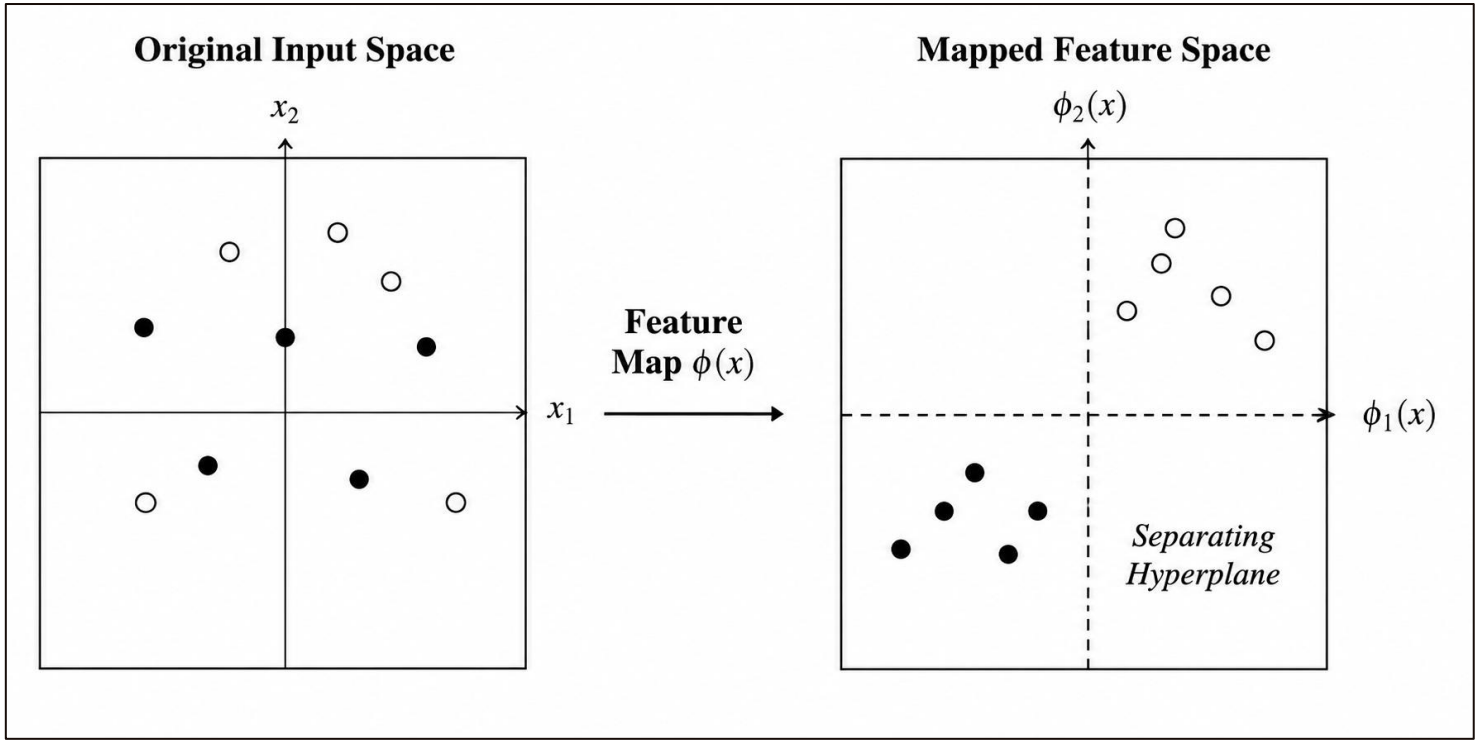
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Feature maps, Kernels and Cover's Theorem

Many learning tasks benefit from transforming the data into a new space where the structure becomes more **amenable to linear separation**.

Mathematical basis, **Cover theorem:**

"A non-linearly separable dataset in a low-dimensional space is highly likely to become linearly separable when mapped non-linearly into a higher-dimensional space"



This transformation is performed by a feature map: $\phi : \mathbb{R}^d \rightarrow \mathcal{F}$, (dimension D>>d).

A simple linear model in feature space is then given by: $h(x) = \langle w, \varphi(x) \rangle + b$ where $w \in \mathcal{F}$ and $b \in \mathbb{R}$.

If ϕ(x) is appropriately chosen, **even a linear model** can capture **nonlinear patterns** in the input space.

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Classical Feature Map and Dual Formulation

$$h(x) = \langle w, \phi(x) \rangle + b$$

The **weight vector w** lives in this infinite- or exponential-dimensional space.

Thanks to the Representer theorem we have the **dual formulation** :

$$h(x) = \left\langle \sum_{i=1}^{M} \alpha_i \phi(x_i) \ , \ \phi(x) \right\rangle + b = \sum_{i=1}^{M} \alpha_i \underbrace{\langle \phi(x_i), \phi(x) \rangle}_{\kappa(x_i, x)} + b$$

$$w = \sum_{i=1}^{M} \alpha_i \phi(x_i)$$

scalar coefficients learned during training

RKHS Kernel function : $k(x, x') = \langle \phi(x), \phi(x') \rangle$ ➡ $h(x) = \sum_{i=1}^{M} \alpha_i k(x_i, x) + b$

To **classify a new data point x,** you never need to compute its explicit coordinates in the massive feature space ϕ(x): you only need to evaluate the kernel value $k(x_i, x)$ between your test point x and the training points x_i . This theorem allows us to bypass the **primal problem** (searching for a vector w in high dimensional space) and solve the dual problem instead (searching form coefficient α_i in a finite space (dataset training space) : this is the "**Kernel Trick**".

Training dataset : $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{M}$

$\min_w \frac{1}{M} \sum_{i=1}^{M} L(h(x_i), y_i) + g(\|w\|^2)$ = Cost function with *Ridge regularisation* $g(\|w\|^2) = \lambda \|w\|^2$

Gram Matrix : $\kappa(x_m, x_l) = K_{ml} \in \mathbb{R}^{M \times M}$ ➡ Optimal weight : $w^* = \sum_{i=1}^{M} \alpha_i \phi(x_i)$

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QSVM vs QRC

From Classical to Quantum kernel

Classical kernel theory (Mercer theorem): To guarantee they will find a global optimum (optimization landscape remains **convex**), the kernel must be positive semi-definite (**PSD**) and correspond to a **valid inner product.**

$$K = \begin{pmatrix} k(x_1,x_1) & k(x_1,x_2) & \dots & k(x_1,x_M) \\ k(x_2,x_1) & k(x_2,x_2) & \dots & k(x_2,x_M) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_M,x_1) & k(x_M,x_2) & \dots & k(x_M,x_M) \end{pmatrix}$$

SPSD ➡ SVM cost function

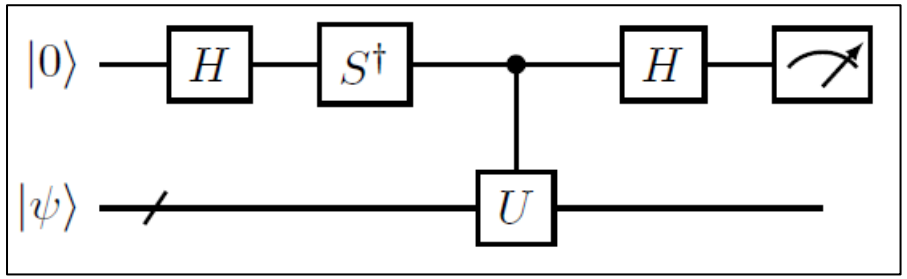
$$\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_M]^T$$

O(M³) complexity :
SVM look for highly qualitative, compact, and diversified training dataset

In the quantum domain, the direct transition amplitude between two different **quantum states** is generally a **complex number:**

$$k(x,x') = \langle \phi(x), \phi(x') \rangle \quad \Rightarrow \quad \langle \psi(x) | \psi(x') \rangle = re^{i\theta}$$

-We cannot directly extract a complex number from a single, standard physical measurement but through a **Hadamard test circuit** (X and Y basis on ancilla qubits).
-Besides the SVM cost function would become a complex non-convex function.



Hadamard test for $\mathrm{Im}\,\langle \psi | U | \psi \rangle$

➡ $$k(x,x') = |\langle \psi(x) | \psi(x') \rangle|^2$$

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From Classical to Quantum kernel

$$\langle \psi(x) | \psi(x') \rangle = re^{i\theta}$$

To transform this complex transition amplitude into a **measurable expectation value**, we use a specific Hermitian operator : the projector onto the state $|\psi(x')\rangle$.

$$A = |\psi(x')\rangle\langle\psi(x')|$$

If we prepare the system in the state $|\psi(x)\rangle$ and measure **the expectation value of this projector** A :

$$\langle A \rangle = \langle \psi(x) | A | \psi(x) \rangle = \langle \psi(x) | \psi(x') \rangle \langle \psi(x') | \psi(x) \rangle = |\langle \psi(x) | \psi(x') \rangle|^2$$

➡ This expectation value is our quantum kernel.

$$k(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$$

The square is the direct consequence of measuring the **expectation value of a projector** which physically corresponds to a **transition probability** or the **fidelity** between the two states.

It looks like a non-linear operation, not a natural inner product but an expectation value <A> is a linear function with respect to the operator A.

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Quantum kernel with density operator

On real quantum hardware, interaction with the environment introduces decoherence, relaxation, and operational noise. Real states are mixed states, mathematically described only by density matrices ρ.

Moving from state vectors to density operators then the quantum kernel is the **Hilbert-Schmidt inner product** between two operators :

$$\rho(x) = |\psi(x)\rangle\langle\psi(x)|$$

$$k(x, x') = \langle \rho(x), \rho(x') \rangle_{\text{HS}} = \text{Tr}\big(\rho(x)\rho(x')\big)$$

$$= \text{Tr}\big(\langle\psi(x')|\psi(x)\rangle \cdot |\psi(x)\rangle\langle\psi(x')|\big)$$

$$= |\langle\psi(x)|\psi(x')\rangle|^2$$

In this operator space, the kernel is strictly linear (Mercer SVM conditions) and the density matrices live in a 4^n *Liouville* space (2^n x 2^n Hermitian matrices vector space)

➡ **Quantum Kernel Estimation** is performing standard linear classification in a massive 4^n dimensional vector space.

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Evaluating Quantum Kernels (SWAP Test)

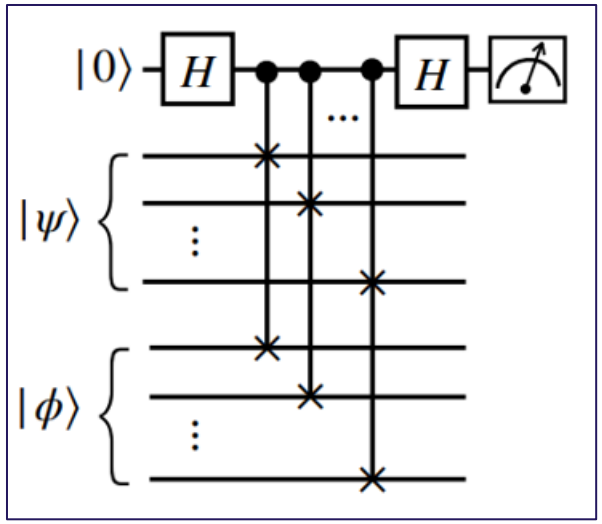
Many machine learning algorithm for tasks like **classification** or regression depends primarily on **inner product between data points** and quantum computing is efficient for this even for high-dimensional vectors.

Overlaps (inner-product) between qubit states, and through global operation which act coherently on the entire vector (classically it iterates over all components).

In practice, the quantum kernel $\kappa(x, x')$ is evaluated using a quantum circuit.
Two common strategies are:

The swap test circuit : A circuit that estimates the overlap $|\langle\phi(x)|\phi(x')\rangle|^2$ by using an ancilla qubit to interfere two quantum states. This requires controlled-swap gates and multiple copies of the states.

Compute-Uncompute (direct fidelity) circuit: If both states $|\phi(x)\rangle$ and $|\phi(x')\rangle$ are prepared on the same device, the fidelity can be estimated by preparing the state $U_\phi(x)^\dagger U_\phi(x')|0\rangle$ and measuring the probability of obtaining the all-zero state.



The swap test

After computing all **pairwise** kernel values over a **training set (M)**, one obtains the kernel matrix K, $K_{ij} = \kappa(x_i, x_j)$.
This matrix is then used as **input** to a **classical kernel** method such as an **SVM**, yielding a hybrid quantum-classical learning pipeline.

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QSVM vs Classical Support Vector Machine (SVM) performance

The **complexity** must be separated into two distinct stages: constructing the problem's geometry (the Gram matrix K) and solving the optimization (calculating the coefficients α_i).

1. Construction of the Gram matrix

The Hilbert space offers an exponentially large feature space natively but QSVM has a fatal flaw for Big Data: to train it, the QPU must physically **cross-compare** every single data point **two-by-two** using resource-intensive SWAP tests. We repeat S times to counter the **shot noise** $O(M^2.S)$

2. Cost of calculating the coefficients α_i

Does **not** take place on the quantum computer. Once the matrix K has been measured is fed into a classical quadratic optimizer $O(M^3)$.

Note: an FTQC variant could use QSVM to solves the optimization directly on the QPU in $O(\log M)$

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QSVM advantages

The advantage of QSVM is not quantitative (speedup), it is qualitative (**expressivity**).

- **SVM** : a major study by the ***Google Quantum AI*** team (Huang, Broughton, Mohseni et al.2021) rigorously demonstrated that even though **classical kernel** (RBF) have infinite dimension they suffer from what is called a "prediction limitation": it requires an exponential amount of training data to successfully map the decision boundary, because its "smooth" inductive bias (e.g. isotropic) is incapable of guessing the underlying structure.

- **QSVM** projects data into a 2^n dimensional Hilbert space via a non-linear quantum feature map.

  It is finite but orientable in geometric directions that are completely inaccessible to classical computation. The unique advantage of QSVM therefore lies in its ability to **discover patterns** invisible to classical algorithms, thereby allowing for better geometric accuracy or solving classification problems where a classical SVM stagnates

The transition from QSVM to QRC

In QSVM, the similarity between these two states is evaluated globally using the quantum kernel function is typically measured on a QPU using a *SWAP test* or a *Compute-Uncompute* circuit :

$$K(X,Y) = \mathrm{Tr}\big(\rho(t)\rho(t')\big)$$

To bridge this to the **reservoir readout**, we introduce the complete orthonormal Pauli basis for the vector space of 2^n x 2^n complex matrices (like density matrix) with the Hilbert-Schmidt inner product :

$$\mu_i = \frac{1}{\sqrt{2^N}}\big(\sigma_{i_1} \otimes \sigma_{i_2} \otimes \cdots \otimes \sigma_{i_N}\big) \quad \text{where } \sigma \in \{I, X, Y, Z\} \qquad \mathrm{Tr}(\mu_i \mu_j) = \delta_{ij}$$

$$\rho(t) = \sum_{i=1}^{4^N} c_i(t)\mu_i$$

The classical coefficients $c_i(t)$ are exactly the **expectation values of our quantum observables** :

$$c_i(t) = \mathrm{Tr}\big(\rho(t)\mu_i\big) = \langle \mu_i \rangle_t \quad \rightarrow \quad \mathrm{Tr}\big(\rho(t)\rho(t')\big) = \sum_{i=1}^{4^N} \langle \mu_i \rangle_t \langle \mu_i \rangle_{t'} \quad \rightarrow \quad K_{\mathrm{QRC}}(t,t') = \sum_{i \in \mathcal{S}} \langle \mu_i \rangle_t \langle \mu_i \rangle_{t'}$$

This **truncation** is an algorithmic advantage : QSVM, we treat all regions of the Hilbert space equally—including pure noise. In QRC, restricting the subset S to a **polynomial-sized of local observables**
(e.g., weight-1 and weight-2 Pauli strings like Z_i or Z_i Z_j) (even though this inner product matrix K_{QRC} is not explicitly calculated)

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The transition from QSVM to QRC

This amounts exactly to Feature Selection or **Kernel Alignment**, just as it is done in advanced QSVM which choose the axes of the Hilbert space contain the physics of the problem (symmetries) and ignore the rest.

QRC does not abandon the power of quantum kernels (QSVM) it makes it scalable. By locally measuring the observables in linear time O(M) **we export the kernel structure into the classical linear Ridge regression**, implicitly navigating the exact same expressive landscape as a Truncated Quantum Kernel.

- In QSVM, the data size M is the bottleneck : it scales quadratically on the QPU and up to cubically on the CPU.
- In QRC, the data size M only impacts the timeline linearly *O*(M).

Unifying QRC and Quantum Kernel Theory

In conclusion,

We have anchored QRC within the formal **framework of kernel methods**—as established by **Maria Schuld**—fundamentally elevates the architecture from a physical heuristic to a rigorous, mathematically sound paradigm.
(“**Supervised quantum machine learning models are kernel methods (2021)”** - https://arxiv.org/abs/2101.11020)

QRC by reading out a controlled number of local observables or classical shadows, it bypasses the data-loading and measurement bottlenecks entirely.

This collapses the **computational complexity** to a strictly **linear scaling** with respect to the dataset size. QRC elegantly reconciles the vast expressivity of exponential Hilbert spaces with the constraints of large-scale classical data processing, providing a scalable, robust, and theoretically grounded path forward for practical Quantum Machine Learning.

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QSVM Applications Target Benchmark

Application	Use Case	Methods
Network Intrusion Detection for Cyberdefense	Automatic detection of network attacks from data streams.	**QSVM** with quantum kernels for supervised classification.
Anomaly Detection in Cybersecurity Audits	Identification of anomalies in high-dimensional audit logs.	**QSVM**, QNN, and QPCA for analysis and dimensionality reduction.
Radar Classification of Airborne Targets	Identification of airborne targets via micro-Doppler radar signatures.	**QSVM** applied to radar signatures.
Satellite and Hyperspectral Image Classification for Intelligence, Surveillance, and Reconnaissance	Satellite image analysis for recognition and mapping.	**QSVM** and quantum kernels for image classification.

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Fine-Tuning of the Reservoir

Angle Encoding

The core idea of angle encoding is to embed classical data into a quantum state through rotation angles.

$$\boldsymbol{x} = (x_0, \ldots, x_i, \ldots, x_{N-1}) \in \mathbb{R}^N$$

the encoded quantum state can be represented as :

$$|\psi\rangle = \bigotimes_{i=0}^{N-1} R_\sigma(x_i)|0\rangle^{\otimes N} = \bigotimes_{i=0}^{N-1} \exp\left(-i\frac{x_i}{2}\sigma\right)|0\rangle^{\otimes N}$$

where $\sigma \in \{X, Y, Z\}$

- To ensure mathematical injectivity it is essential to scale each element into the range $[0, \pi)$ to ensure that different values are encoded into distinct quantum states.

- A key advantage of angle encoding is its ability to **introduce nonlinearity**. By mapping classical data into the parameters of quantum rotation gates, angle encoding uses **trigonometric functions** to naturally capture nonlinear relationships.

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Angle Encoding & Re-uploading: unlocking the Fourier Spectrum

Quantum model 's expressivity via Trigonometric Polynomials

Quantum model : $f(x) = \langle 0|U^\dagger(x,\theta)OU(x,\theta)\,|0\rangle = \sum_{\omega\in\Omega} c_\omega e^{i\omega x}, \quad \text{frequency } \omega = \mu_{\mathbf{k}} - \mu_{\mathbf{j}},$

We use the Hamiltonian encoding via gates of the form $G(x) = e^{-ixH}$ $\quad$ eigenvalues $\lambda_1, \ldots, \lambda_d$.

The degree of the spectrum $D = \max(\Omega)$ $\quad$ $\Omega = \{\mu_{\mathbf{k}} - \mu_{\mathbf{j}} \mid \mathbf{k}, \mathbf{j} \in [d]^L\}$,

Two independent factors for the expressivity :

- the structure of the **frequency spectrum** Ω governed by the eigenvalues of the **encoding Hamiltonians**,
- the **controllability** of the coefficients c_ω which depends on the (fixed) **unitaries** and **measurement** observable.

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Parallel repetition of Pauli encodings across r qubits

$$S(x) = e^{-i\frac{x}{2}\sigma_r} \otimes \cdots \otimes e^{-i\frac{x}{2}\sigma_1}, \quad \sigma_j \in \{\sigma_x, \sigma_y, \sigma_z\} \quad \rightarrow \quad S(x) = V^\dagger \exp\left(-i\frac{x}{2}\sum_{q=1}^{r}\sigma_z^{(q)}\right) V = V^\dagger e^{-ix\Lambda} V,$$

Since these operators act on independent Hilbert spaces, they all commute with one another. The global diagonalized operator is the sum of the individual Hamiltonians of each qubit :

$$\Lambda = \frac{1}{2}\sum_{q=1}^{r}\sigma_z^{(q)}$$

The eigenvalues of a sum of independent, commuting operators, the result is simply **the sum of all possible combinations of their individual eigenvalues**.

$$\lambda_p = p\cdot\left(\frac{1}{2}\right) + (r-p)\cdot\left(-\frac{1}{2}\right) = \frac{2p-r}{2} \qquad p \in \{0,1,\ldots,r\}$$

P = the number of qubits which takes eigenvalue +1/2

The Fourier formalism tells us that the accessible frequencies w are the difference between any two eigenvalues of the system.

$$\omega = \left(\frac{2p-r}{2}\right) - \left(\frac{2p'-r}{2}\right) = \frac{2p-2p'}{2}$$

$$\omega = p - p' \quad \text{avec } p, p' \in \{0,1,\ldots,r\}$$

$$\Omega_{par} = \{-r, -r+1, \ldots, 0, \ldots, r-1, r\}$$

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Sequential repetition of Pauli encodings across r Layers

$$U(x) = W^{(r+1)} e^{-i\frac{x}{2}\sigma_r} W^{(r)} \dots W^{(2)} e^{-i\frac{x}{2}\sigma_1} W^{(1)}$$

→ H or R_y gate to create a superposition state

Assume all σ_j are equal and act on the same qubit. As before, we diagonalize each encoding gate to get $\Lambda = \frac{1}{2}\sigma_z$, and the full spectrum consists of differences of sums of r eigenvalues:

According to the general framework the quantum state after passing through L=r layers encoding layers is a sum over all possible index trajectories:

$$[U(x)|0\rangle]_i = \sum_{j \in [2]^r} e^{-i\mu_j x} \mathcal{W}_j$$

- $j = (j_1, j_2, \dots, j_r)$ is a multi-index describing the "path" of eigenvalues chosen at each layer l. Since $d = 2$, each index j_l belongs to the set $\{1, 2\}$.
- μ_j is the total phase accumulated along path j, which is the sum of the eigenvalues encountered at each layer:

$$\mu_j = \sum_{l=1}^{r} \lambda_{j_l}$$

The output function is a double sum over all possible path combinations : $f(x) = \sum_{k,j \in [2]^r} e^{i(\mu_k - \mu_j)x} a_{k,j}$

The frequencies accessible by the model, arise from the phase differences of these paths: $\omega = \mu_k - \mu_j$

$$\Omega_{\text{seq}} = \left\{ \left(\sum_{l=1}^{r} \lambda_{k_l} \right) - \left(\sum_{l=1}^{r} \lambda_{j_l} \right) \middle| \, k_l, j_l \in \{1, 2\} \right\} \Rightarrow \Omega_{\text{seq}} = \{-r, -r+1, \dots, r-1, r\}.$$

Thus, the frequency spectrum grows linearly with the number of repetitions r

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How to extend the frequency spectrum accessible to a quantum model ?

The Fourier degree can be increased linearly by **repeating the data-encoding gates (re-uploading)** —either in parallel (within a single layer) or sequentially (across multiple layers) :

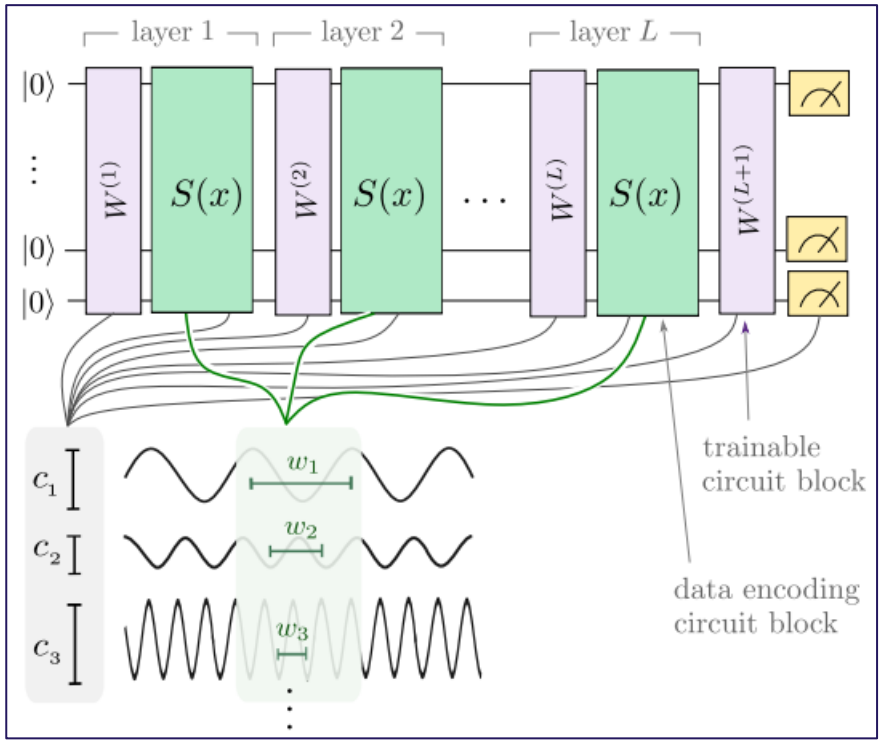
- **Parallel repetition**: A single-layer model (L=1) where the encoding gate is repeated r in parallel across different qubits.
- **Sequential repetition**: A multi-layer model (L =r) where a single-qubit encoding gate is applied repeatedly across r layers. (but with Unitaries W in between !)

➡ $\Omega_{\text{seq}} = \{-r, -r+1, \ldots, r-1, r\}$

Conclusion:

The frequency spectrum grows linearly with the **number of repetitions r**, whether applied in **parallel** or **sequentially** (**truncated Fourier series of degree r**)

….and this scaling mechanism is independent of other details like the choice of fixed **unitaries** (ansatz) or measurement **observable**.



(*Fig.) "The effect of data encoding on the expressive power of QML" https://arxiv.org/pdf/2008.08605 (M. Schuld 2021)

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Exponential Encoding

Parallel encoding operator : $S(x) = \bigotimes_{j=1}^{N} e^{-i\frac{x\beta_j}{2}\sigma^z}$

Computational basis state $|k\rangle = |k_1 k_2 \dots k_N\rangle$

$\sigma^z|0\rangle = +1|0\rangle$

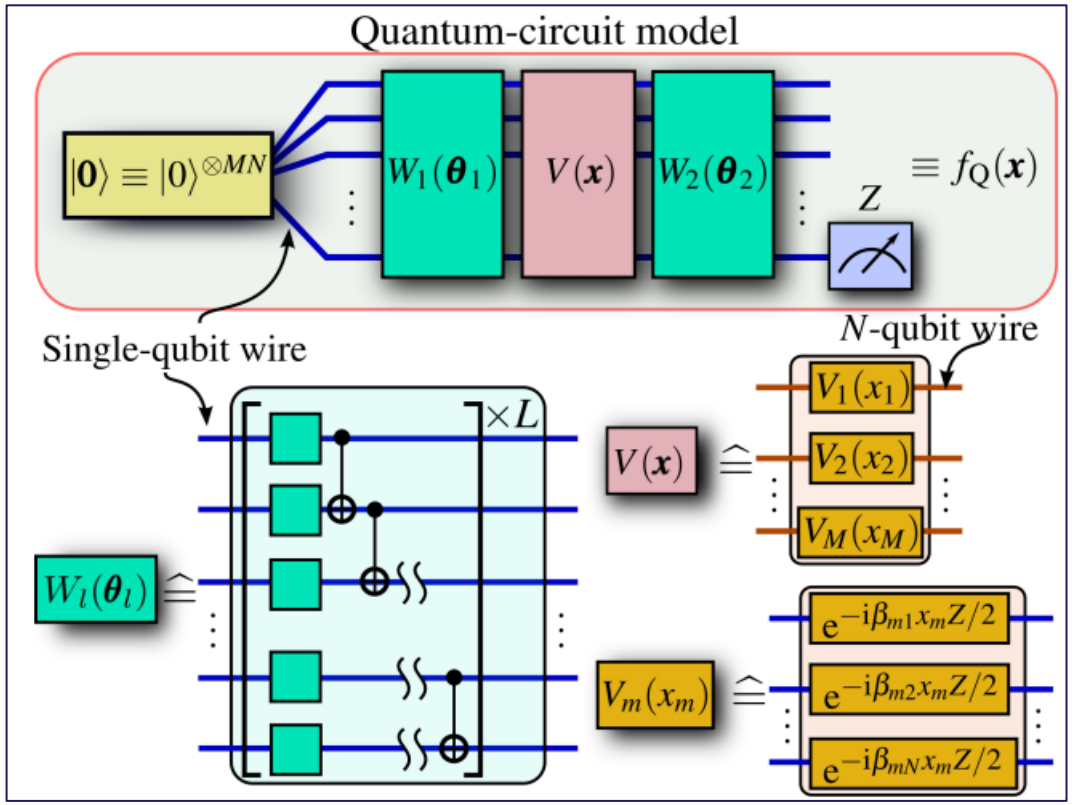
$\sigma^z|1\rangle = -1|1\rangle$

$(-1)^{k_j} = 1 - 2k_j$

$$e^{-i\frac{x\beta_j}{2}\sigma^z}|k_j\rangle = e^{-i\frac{x\beta_j}{2}(1-2k_j)}|k_j\rangle = e^{-i\frac{x\beta_j}{2}} e^{ix\beta_j k_j}|k_j\rangle$$

$$S(x)|k\rangle = \left(\prod_{j=1}^{N} e^{-i\frac{x\beta_j}{2}}\right)\left(\prod_{j=1}^{N} e^{ix\beta_j k_j}\right)|k\rangle$$

$$= e^{-i\frac{x}{2}\sum_{j=1}^{N}\beta_j} e^{ix\sum_{j=1}^{N}\beta_j k_j}|k\rangle$$



Exponential data encoding for quantum supervised learning **https://arxiv.org/abs/2206.12105 (Shin, 2023)**

In supervised QML, the QRC function f(x) is : $f(x) = \langle\psi|S^\dagger(x)OS(x)|\psi\rangle$ Before the encoding block : $|\psi\rangle = \sum_{k\in\{0,1\}^N} c_k|k\rangle$

➡ $$S(x)|\psi\rangle = e^{-i\frac{x}{2}\sum_{j=1}^{N}\beta_j} \sum_{k\in\{0,1\}^N} c_k e^{ix\sum_{j=1}^{N}\beta_j k_j}|k\rangle$$

$$f(x) = \sum_{k,k'} c_{k'}^* c_k \langle k'|O|k\rangle e^{ix\sum_{j=1}^{N}\beta_j k_j} e^{-ix\sum_{j=1}^{N}\beta_j k_j'} = \sum_{k,k'} \mathcal{A}_{k,k'} e^{-ix\sum_{j=1}^{N}(k_j'-k_j)\beta_j}$$

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Extraction of the frequency spectrum

The frequency spectrum of a Fourier series is the set of all numerical coefficients that multiply the input variable x :

$$f(x) = \sum_{k,k'} \mathcal{A}_{k,k'} e^{-ix\sum_{j=1}^{N}(k'_j - k_j)\beta_j} \qquad \mathcal{A}_{k,k'} = c^*_{k'} c_k \langle k'|O|k\rangle$$

The set of the frequency generated by the combination of two basis states $|k\rangle$ and $|k'\rangle$ is: $\omega = \sum_{j=1}^{N}(k'_j - k_j)\beta_j$

The set of all frequencies mathematically accessible by this model is : $\Omega^{(N)} = \left\{ \sum_{j=1}^{N}(k'_j - k_j)\beta_j \;\middle|\; k_j, k'_j \in \{0,1\} \right\}$

$b_j = k'_j - k_j \qquad b_j \in \{-1,0,1\} \quad \Rightarrow \quad \omega = \sum_{j=1}^{N} b_j \beta_j.$

Example of dense , non-degenetated spectrum : $\beta_j = 2\sum_{i=1}^{j-1}\beta_i + 1 = 3^{j-1} \qquad \beta_1 = 1$

$$\Rightarrow \qquad \Omega = \left\{ \sum_{j=1}^{N} b_j 3^{j-1} \;\middle|\; b_j \in \{-1,0,1\} \right\}$$

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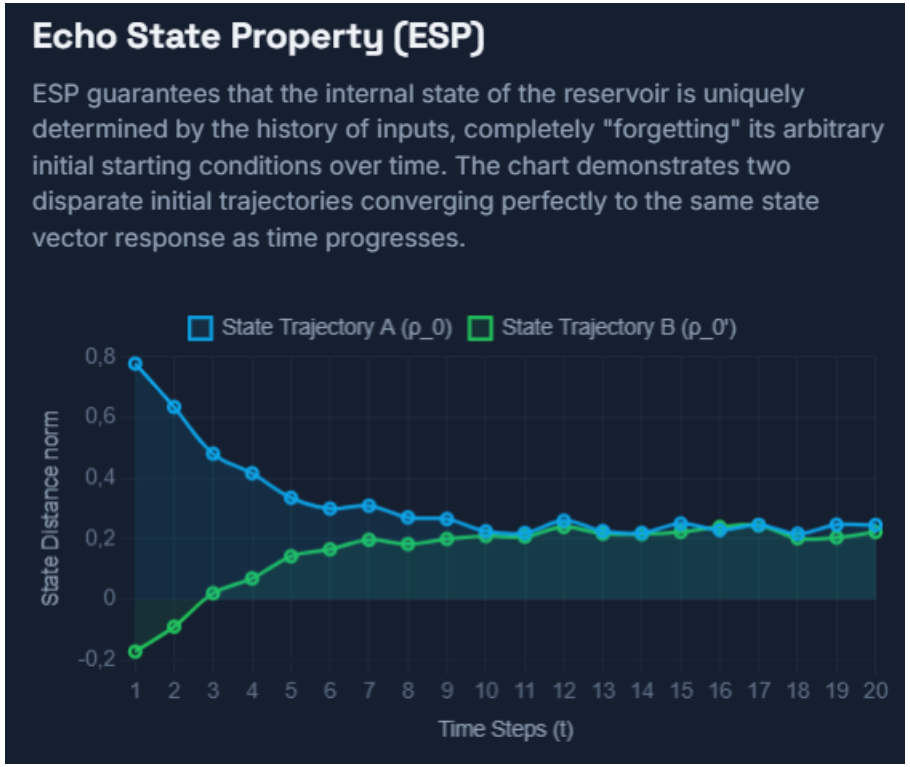
Basic ‘physical’ properties of (Q)RC system

Echo State Property (ESP)

The reservoir state must be a **unique function** of the input history. This ensures that the system "**forgets**" its arbitrary initial conditions to converge toward a stable and reproducible response.

Fading Memory (FM)

The influence of past inputs must gradually fade over time. This property prevents saturation or chaotic behavior, allowing the reservoir to focus on recent temporal correlations.



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QRC for an early FTQC approach

- QRC does not abandon the power of quantum kernels, it makes them scalable.
- **Streaming strategy for Big Data** : QRC is processes data sequentially with a linear complexity. The classical computer only ever needs to invert a fixed matrix of size in the number of measured observables completely independent of the dataset size.
- **Bypassing explicit numerical simulation**: QRC should not be viewed as a physical simulation tool, but rather as a **complex temporal signal processor**.
- **Early-FTQC viability :** purely algorithmic approaches to solving complex equations require millions of fault-tolerant gates and deep, complex oracle structures**.**
- **Eco-energetic quantum Inference:** no back-propagation training required like classical RNN neither Variational Quantum Circuit (VQC) optimization loop avoid Barren plateau problem as well.
- **Exponential Dimensionality** of QRC vs. classical AI : 10^{12} (one trillion) vs 10^{15} for 50 qubits.
- But we must design a structured, problem-specific feature maps because we want to sample only lower-dimensional sub-manifolds of the Hilbert space (to prevent concentration of measure).

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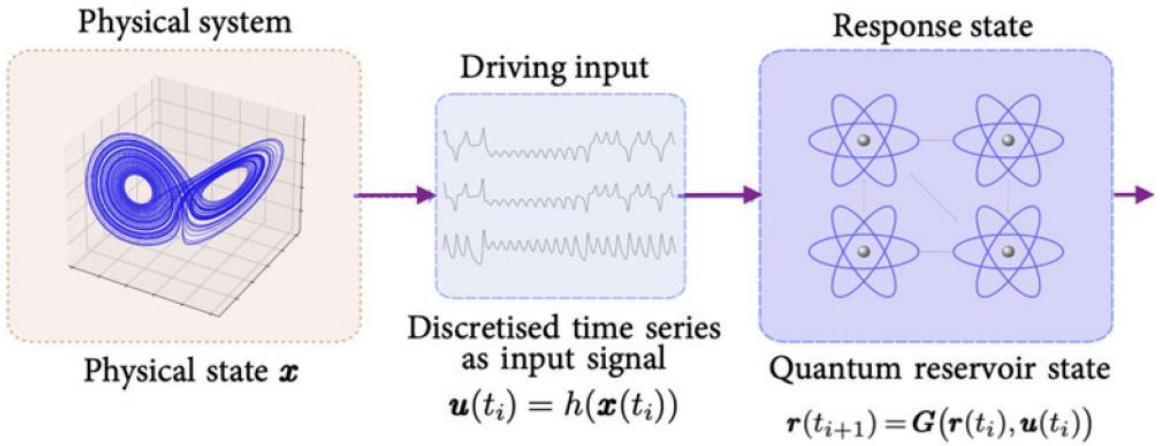
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Scientific papers analysis & Experimental Proofs

Using Lorenz-96 as a Model of Weather Prediction

The Lorenz-96 model simulates atmospheric circulation at mid-latitudes. Mathematically, the model is governed by a set of (40) coupled nonlinear differential equations:

$$\frac{dX_i}{dt} = (X_{i+1} - X_{i-2})X_{i-1} - X_i + F$$

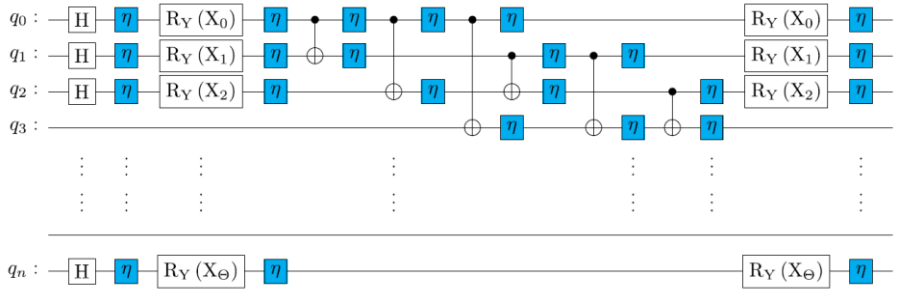


which, under sufficient external forcing (F>=8) yields a **chaotic strange attractor behavior** characterized by a fractal non-integer dimension.

An optimal reservoir "Physics-Informed" QRC.

The Lorenz-96 model is a toy mathematical model that acts as a highly idealized representation of the atmosphere, yet it preserves the essential dynamical invariants necessary for forecasting chaotic dynamics.

By computing the **Jacobian of the QRC** to find the Lyapunov spectrum and Kaplan-Yorke dimension. This allows us to ensure through hyperparameter grid search that it faithfully reconstructs the target Lorenz-96 attractor which represent our mathematical "ground truth“.



Ref : https://doi.org/10.1098/rspa.2025.0550

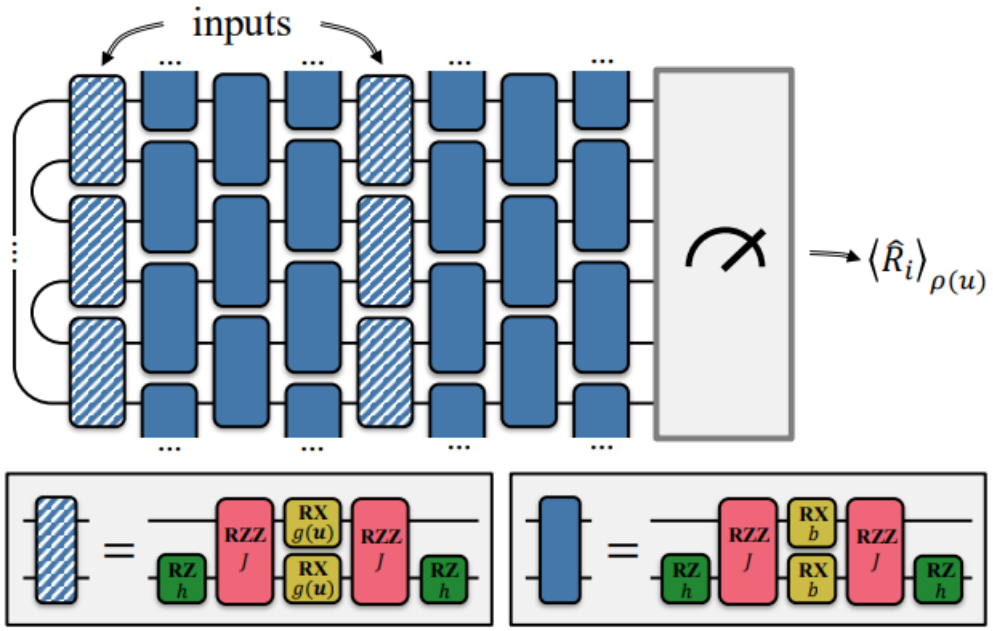
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Breaking concentration barriers for QELM on digital QPU

https://arxiv.org/abs/2603.13005 mars 2026)

Timothée Dao [1,2,*] Ege Yilmaz [3,1,†] Ibrahim Shehzad [4] Christophe Pere,[5] Kumar Ghosh [6] Isabelle Wittmann,[1] Thomas Brunschwiler,[1] Giorgio Cortiana [6] Corey O'Meara [6] Stefan Woerner [1] and Francesco Tacchino [1,‡]

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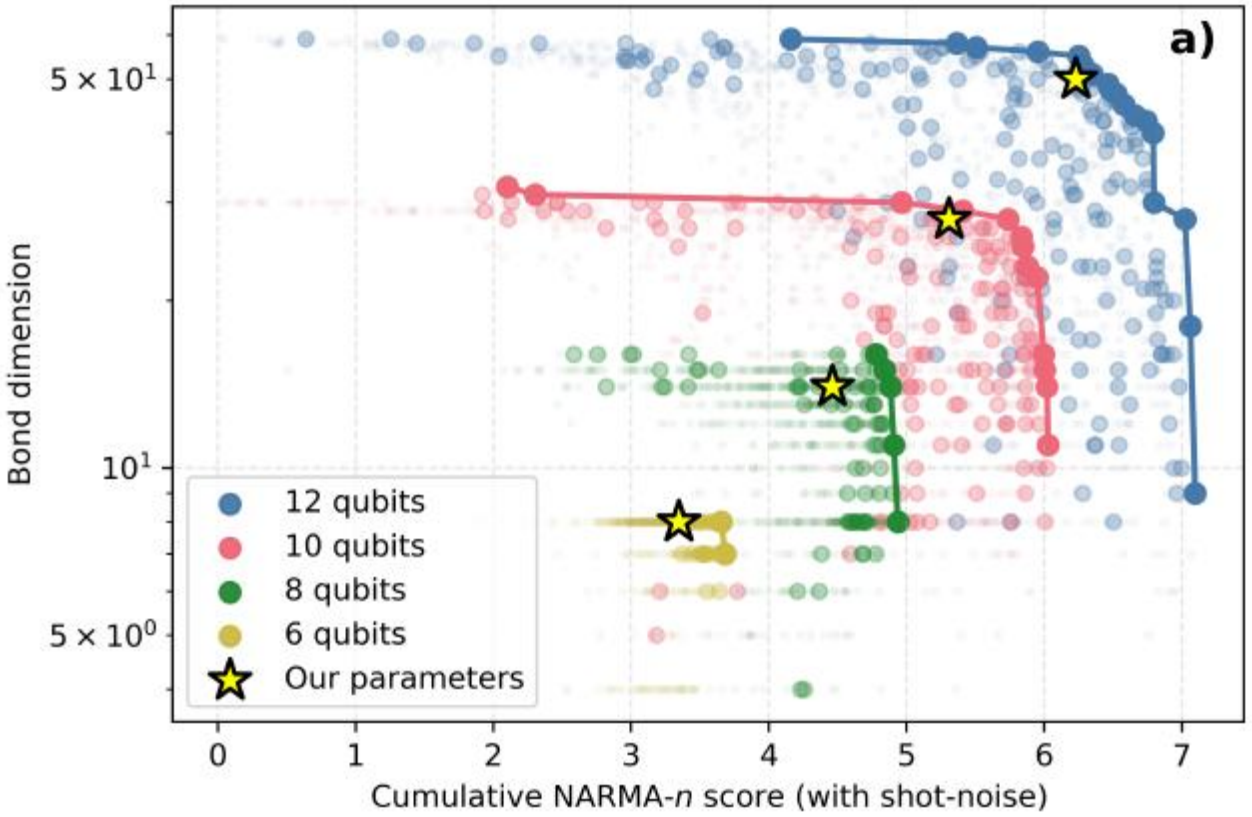
$$\rho(\mathbf{u}) = \left(\prod_{j=1}^{\ell} U_{\mathrm{RES}} U^{(j)}(\mathbf{u}) \right) \rho_0 \left(\prod_{j=1}^{\ell} U_{\mathrm{RES}} U^{(j)}(\mathbf{u}) \right)^{\dagger}$$

$$\hat{H}_0 = J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1} + h \sum_{n=0}^{N-1} \hat{Z}_i \quad \text{and} \quad \hat{H}_K = b \sum_{n=0}^{N-1} \hat{X}_i$$

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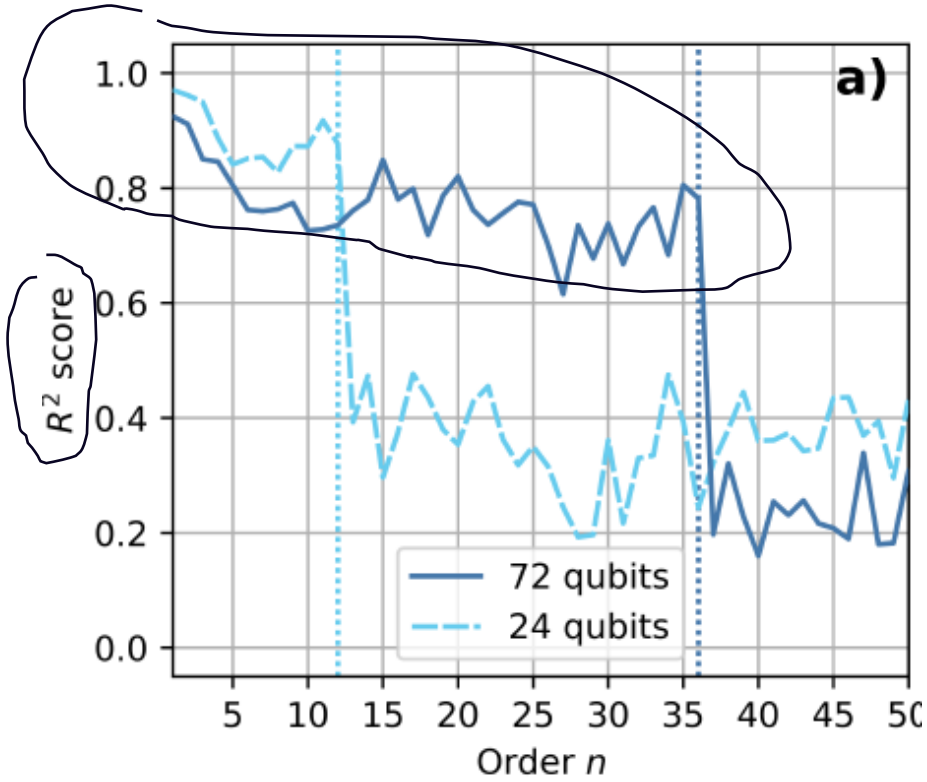
Quantum Extreme Learning Machine on IBM Heron 124 qubits

Objective : To find a "**universal operational regime**," thereby validating the balance between noise resilience, entanglement complexity, and variability.



The empirical evidence in this figure demonstrates that hyperparameters optimized at a small scale (8 to 12 qubits) lie directly on the **Pareto front** for 124-qubit systems

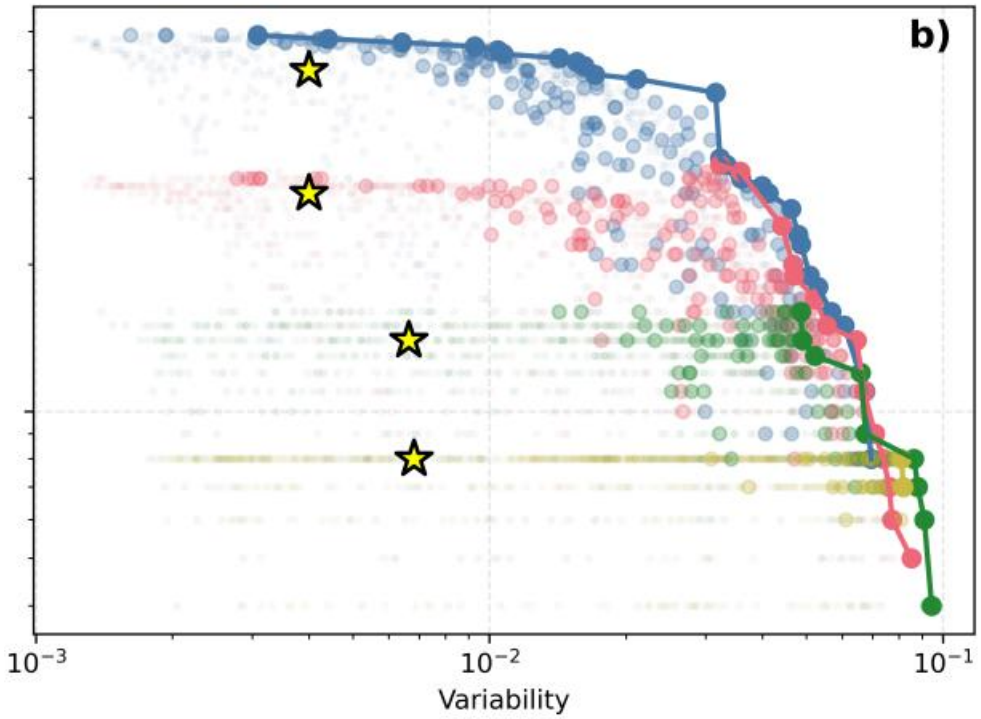
Increasing the number of qubits directly extends the effective **memory horizon** from n = 24 à 72 qubits.



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Emergence of a universal operating regime up to 124 qubits

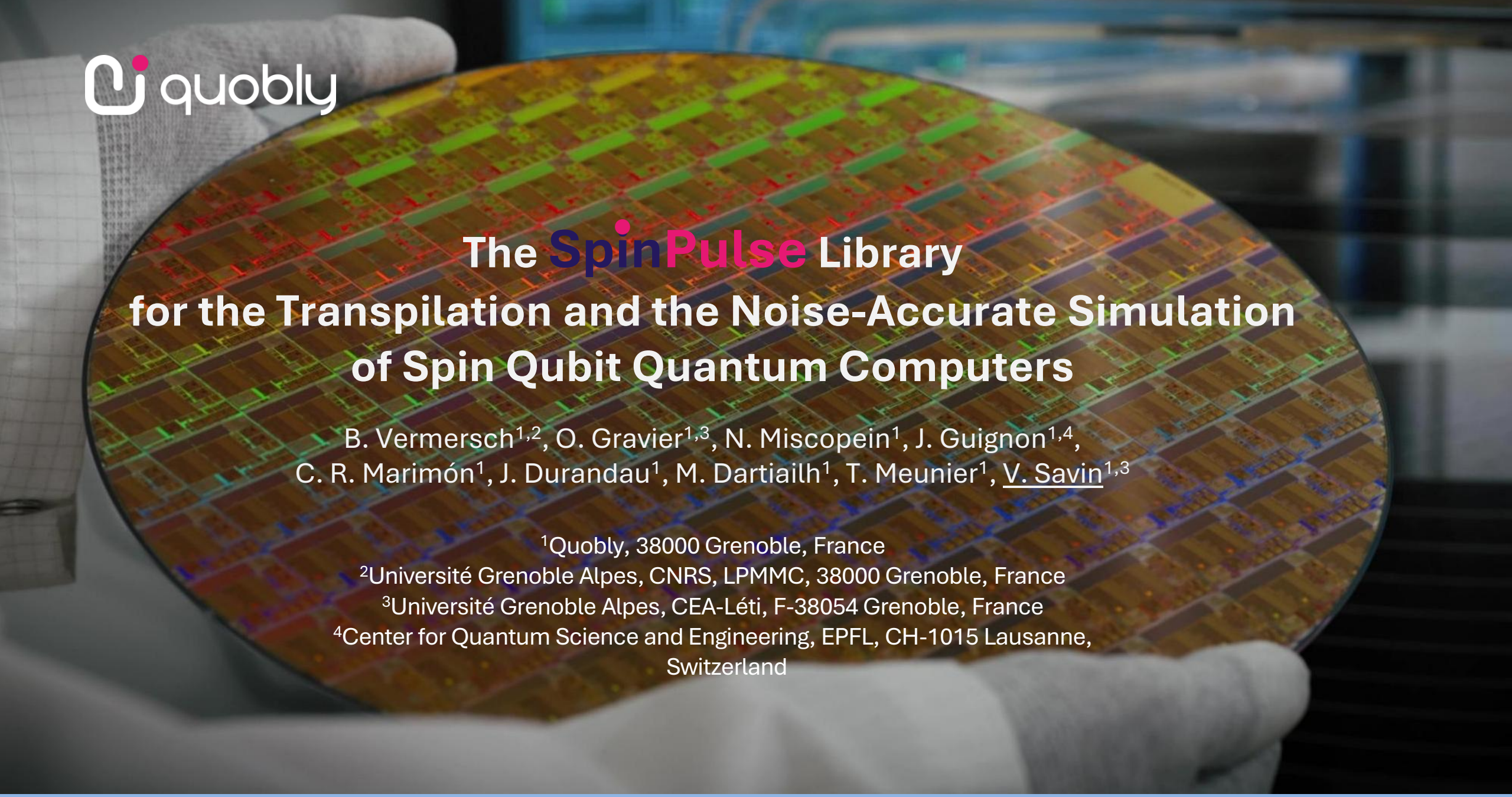
3rd metric on the Pareto Front



Trade-off between observable variability and bond dimension, showing that the selected operating point occupies a regime of high entanglement richness while preserving sufficient output variability to avoid over-concentration.

Together, these projections highlight the emergence of a universal operating regime that generalizes well across architectures

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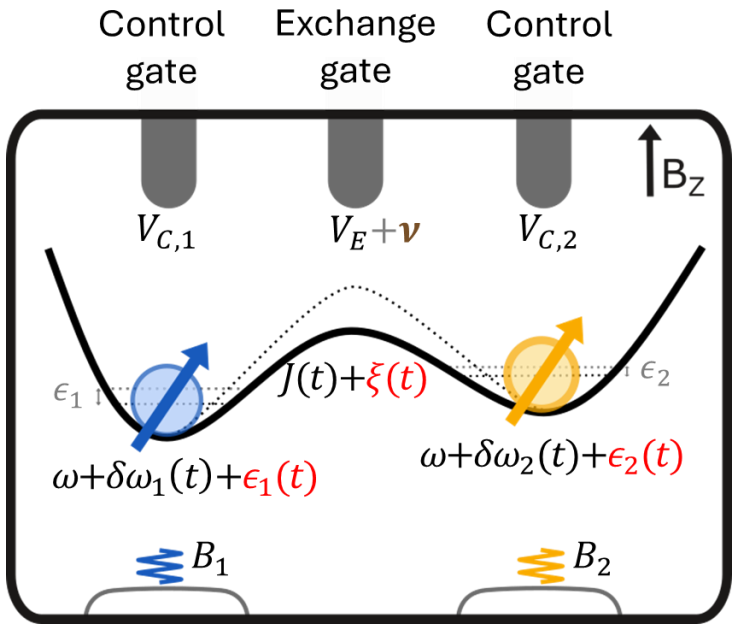


The SpinPulse Library for the Transpilation and the Noise-Accurate Simulation of Spin Qubit Quantum Computers

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Quobly's Hardware Model



- ω: Larmor frequency at rest
- $\delta\omega_i$: DC-generated Stark shift
- B_i: driving field
- J: coupling
- ϵ_i: Larmor frequency noise
- ξ: coupling noise

Single-qubit gates: RZ(θ), RX(θ), RY(θ)

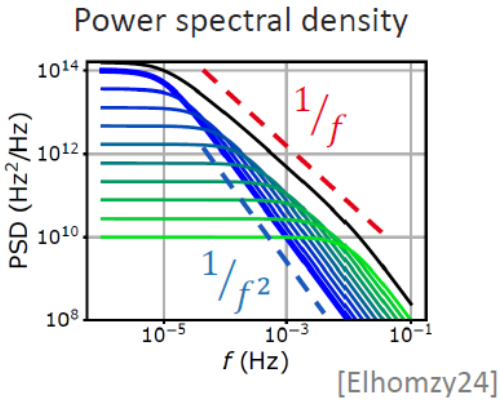
$$h_i(t) = \frac{\delta\omega_i(t)+\epsilon_i(t)}{2} Z_i + \frac{B_i(t)}{2}(\cos(\varphi_i)\, X_i + \sin(\varphi_i)\, Y_i)$$

Two-qubit gates: RZZ(θ)

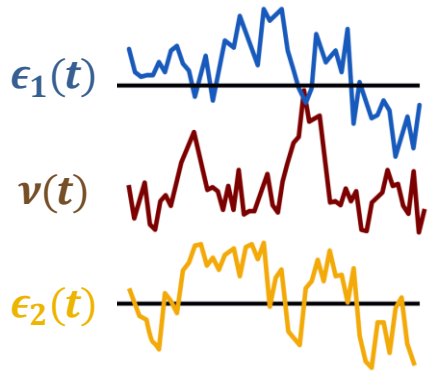
$$h(t) = \frac{J(t)+\xi(t)}{2}(X_1X_2 + Y_1Y_2 + Z_1Z_2) + \frac{\delta\omega_1(t)+\epsilon_1(t)}{2} Z_1 + \frac{\delta\omega_2(t)+\epsilon_2(t)}{2} Z_2$$

Noise models

- Non-Markovian pink noise (PSD $\propto \frac{1}{f^{\alpha}}$)



[Elhomzy24]



SpinPulse

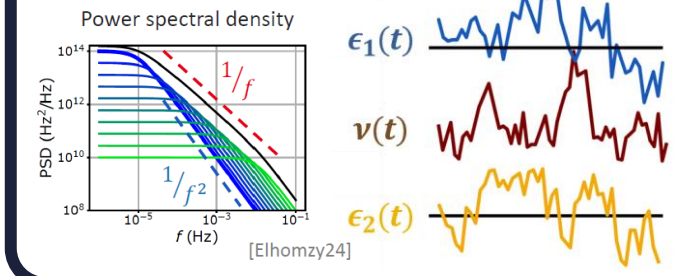
Preprint:
https://arxiv.org/pdf/2601.10435

Doc:
https://quobly-sw.github.io/SpinPulse/

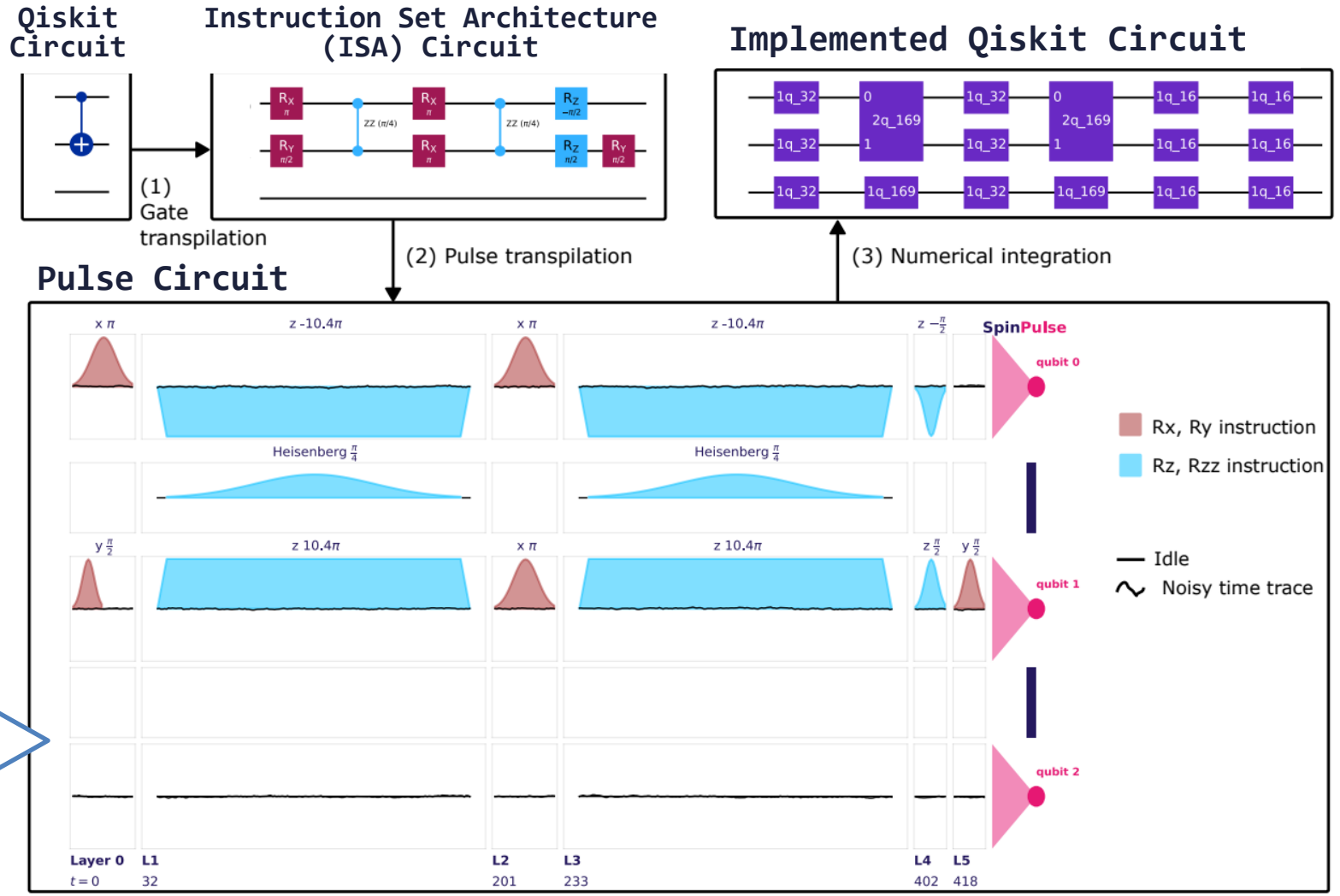
Github:
https://github.com/quobly-sw/SpinPulse

$$h_i(t) = \frac{\delta\omega_i(t)+\epsilon_i(t)}{2} Z_i + \frac{B_i(t)}{2}(\cos(\varphi_i) X_i + \sin(\varphi_i) Y_i)$$

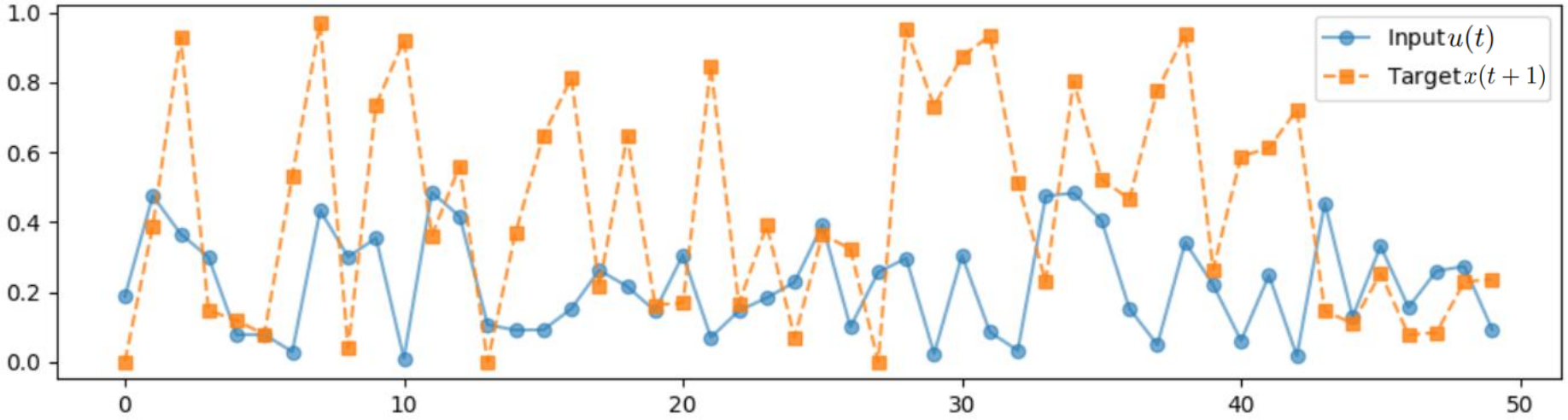
$$h(t) = \frac{J(t)+\xi(t)}{2}(X_1X_2 + Y_1Y_2 + Z_1Z_2) + \frac{\delta\omega_1(t)+\epsilon_1(t)}{2} Z_1 + \frac{\delta\omega_2(t)+\epsilon_2(t)}{2} Z_2$$



[Elhomzy24]



Benchmark Time Series Forecasting: NARMA-N



$$x(t+1) = \alpha x(t) + \beta x(t) \left(\sum_{i=0}^{N-1} x(t-i) \right) + \gamma u(t-N+1)u(t) + \delta$$

Mémoire & Non-linéarité

Le benchmark NARMA (Nonlinear Auto-Regressive Moving Average) est le standard d'or pour évaluer les performances d'un réservoir quantique.

- **Profondeur de mémoire (La Somme):** Teste la capacité du système à retenir des informations sur **n pas de temps**.
- **Transformation complexe** : le terme "Non-Linéaire u^2“ : C'est le plus dur. Le système doit multiplier l'entrée actuelle par une entrée reçue il y a longtemps (t-n). Cela demande au réservoir de mélanger physiquement des informations séparées dans le temps.
- **Standard industriel :** Utilisé pour comparer la capacité de traitement d'information (IPC) de différents Hamiltoniens.

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Coefficient de Détermination (R^2)

Indicateur de précision du Reservoir Computing sur les séries temporelles

Coefficient/Score Global

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Représente la part **de variance du signal expliquée par le modèle réservoir**.

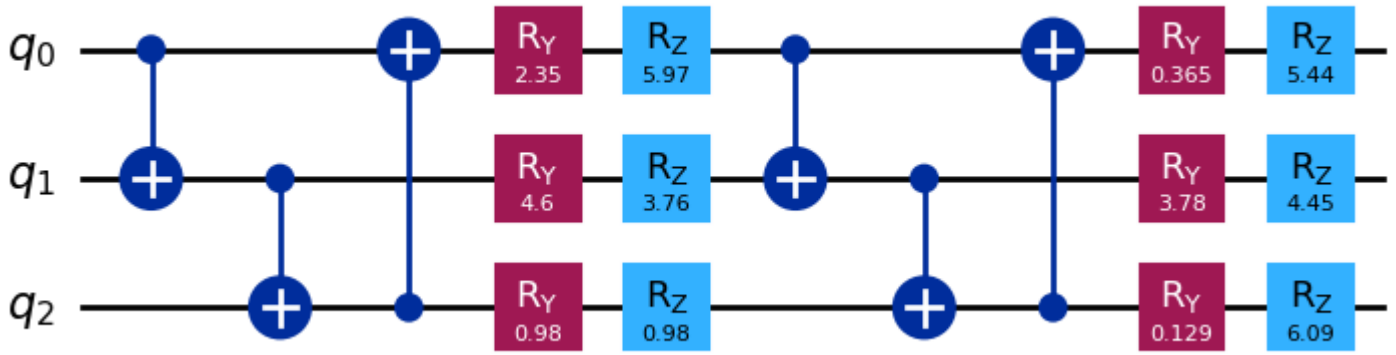
- y_i représente les valeurs observées.
- $\hat{y}_i$ représente les valeurs prédites par votre modèle
- $\bar{y}$ est la moyenne des valeurs observées

Ce score est crucial car il inclut les effets du **shot noise** (bruit d'échantillonnage) et donc l'effet de concentration des mesures.

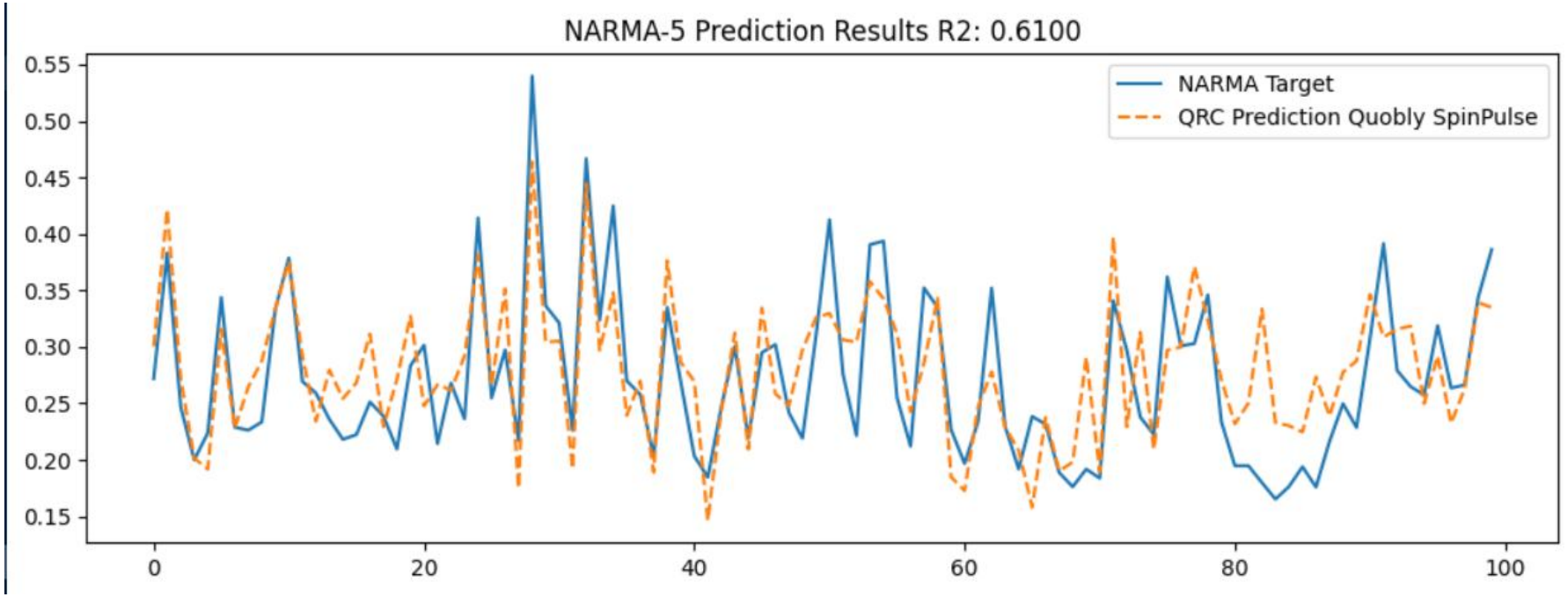
1.0	**Modèle Idéal** Prédiction parfaite. Aucune erreur résiduelle (SSres = 0).
0.7	**Excellent (Seuil NARMA)** La dynamique complexe et non-linéaire est capturée à 70%.
0.0	**Modèle Nul** Le réservoir ne prédit pas mieux qu'une simple ligne moyenne.
< 0	**Modèle Défaillant** Le modèle est pire que la moyenne (désynchronisation ou washout absent).

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NARMA-5 benchmark with SpinPulse



A simple circuit (w/o Encoding part) for a naive QELM (with re-uploading) approach without any optimization



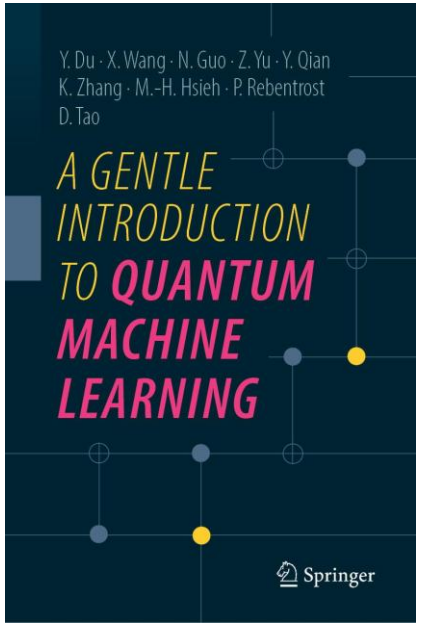
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Open questions & Research directions

Despite this progress, fundamental questions remain open:

- and encoding strategies are best matched to a given signal class ?
- QSVM → QRC migrations with a fundamental link: quantum kernels ↔ quantum reservoir
- Can QOS algorithm help for the encoding strategy ?
- How does noise interact with reservoir expressivity ?

We are backed by a stellar scientific advisory committee





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Thank you for
your attention

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